

Introduction to
Artificial Intelligence
with Python

Language

Natural Language Processing

Natural Language Processing

- automatic summarization
- information extraction
- machine translation
- question answering
- text classification
- ...

Syntax

"Just before nine o'clock Sherlock Holmes stepped briskly into the room."

"Just before Sherlock Holmes nine o'clock stepped briskly the room."

"I saw the man on the mountain
with a telescope."

Semantics

"Just before nine o'clock Sherlock Holmes stepped briskly into the room."

"A few minutes before nine, Sherlock Holmes walked quickly into the room."

"Colorless green ideas sleep furiously."

Natural Language Processing

formal grammar

a system of rules for generating sentences
in a language

Context-Free Grammar

she

saw

the

city

N
|
she

V
|
saw

D
|
the

N
|
city

N → she | city | car | Harry | ...

D → the | a | an | ...

V → saw | ate | walked | ...

P → to | on | over | ...

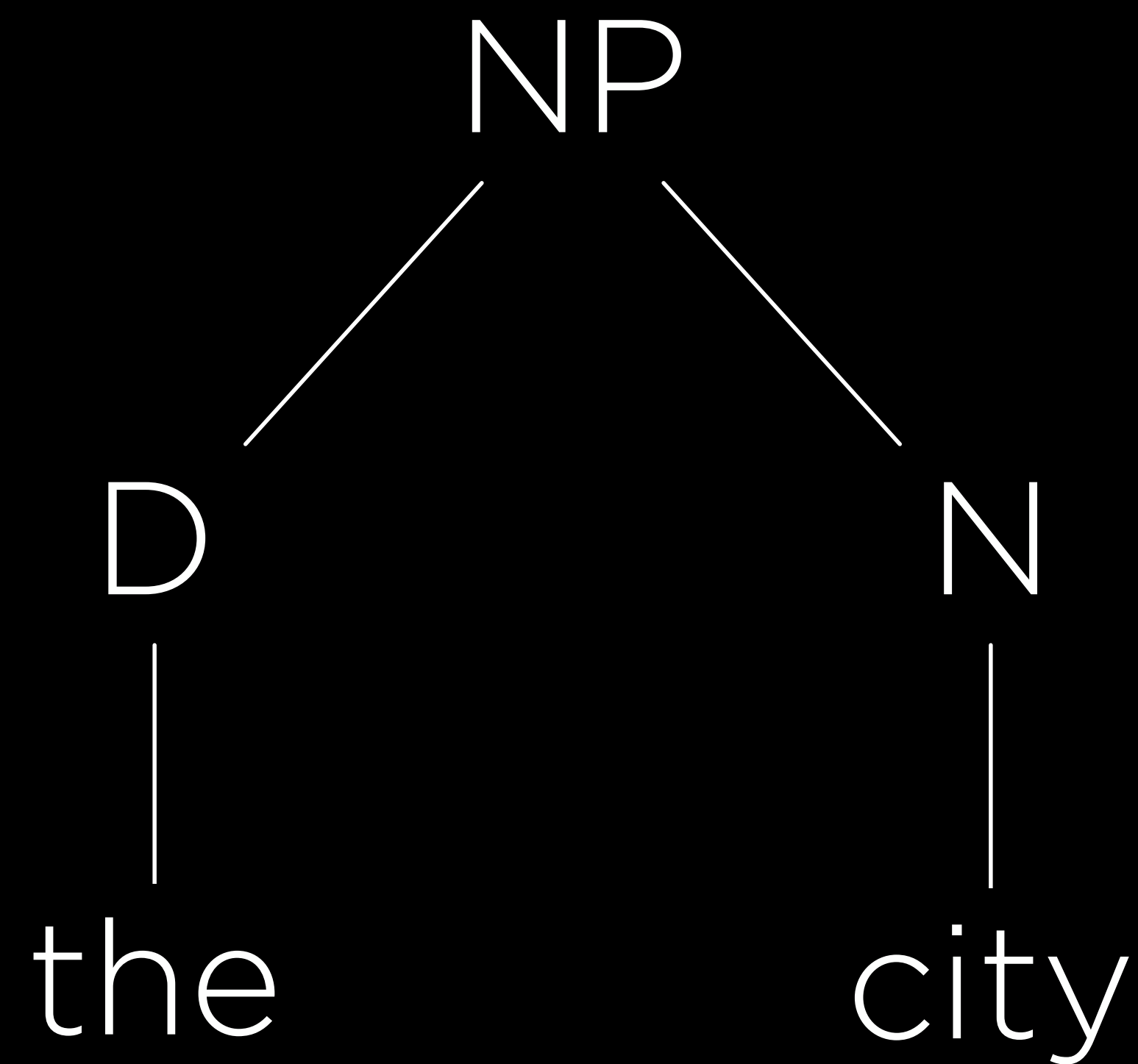
ADJ → blue | busy | old | ...

$NP \rightarrow N \mid D N$

NP → N | D N

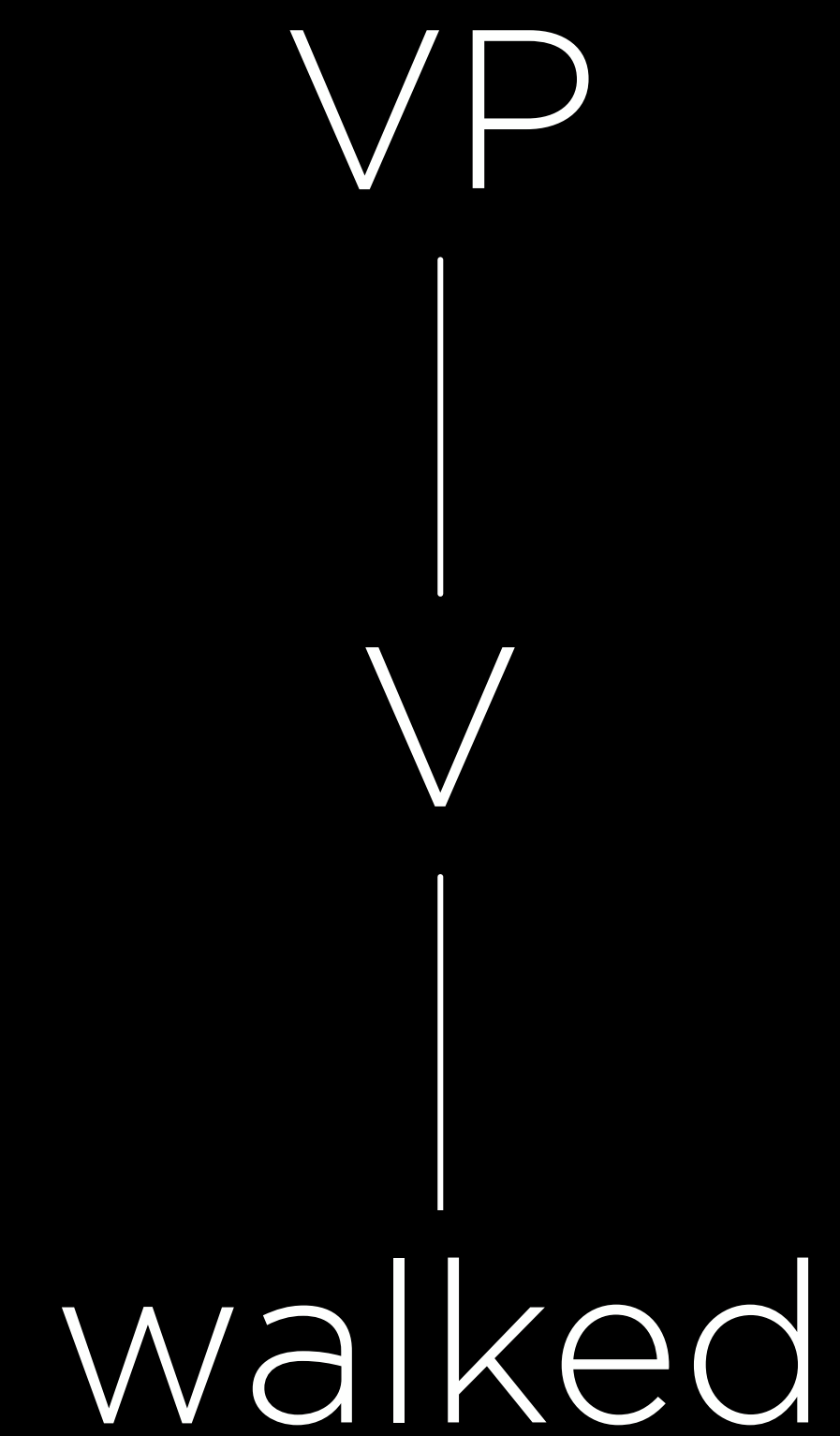
NP
|
N
|
she

NP → N | D N

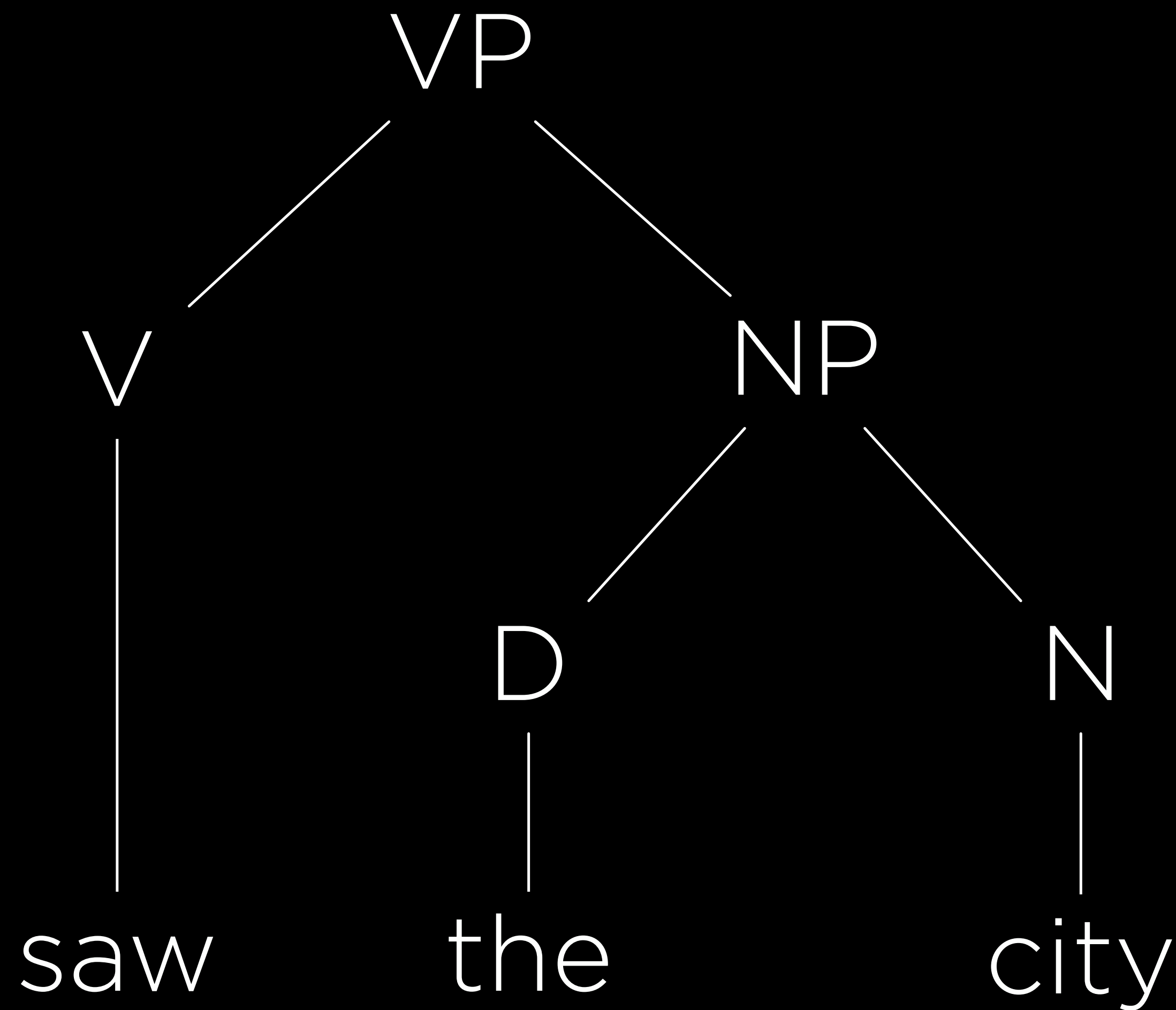


$VP \rightarrow V \mid V NP$

$VP \rightarrow V \mid V NP$

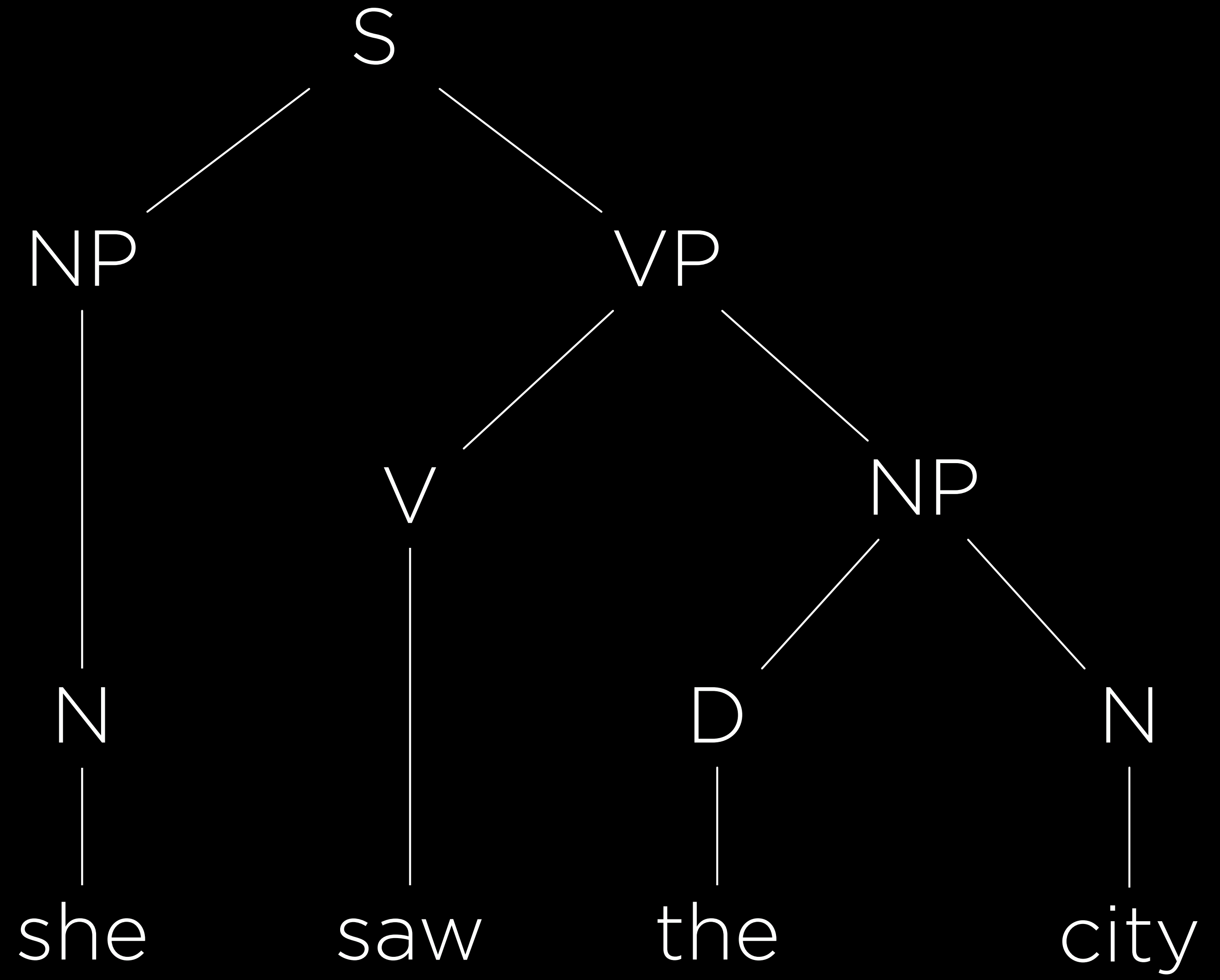


$VP \rightarrow V \mid V NP$



$S \rightarrow NP VP$

$S \rightarrow NP VP$



nlTK

n-gram

a contiguous sequence of *n* items
from a sample of text

"How often have I said to you that when you have eliminated the impossible whatever remains, however improbable, must be the truth?"

"How often have I said to you that when you have eliminated the impossible whatever remains, however improbable, must be the truth?"

"How **often have I** said to you that when you have eliminated the impossible whatever remains, however improbable, must be the truth?"

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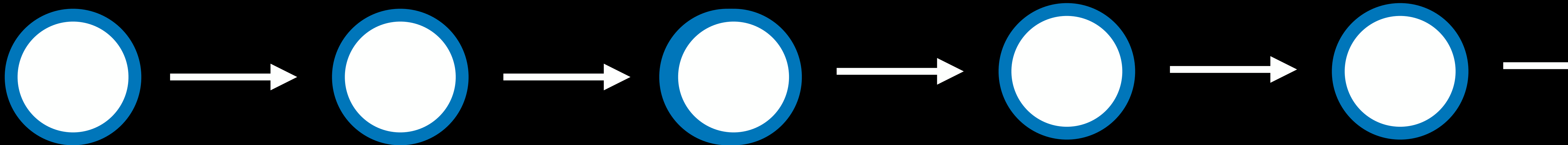
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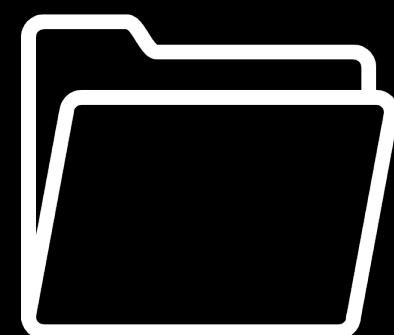
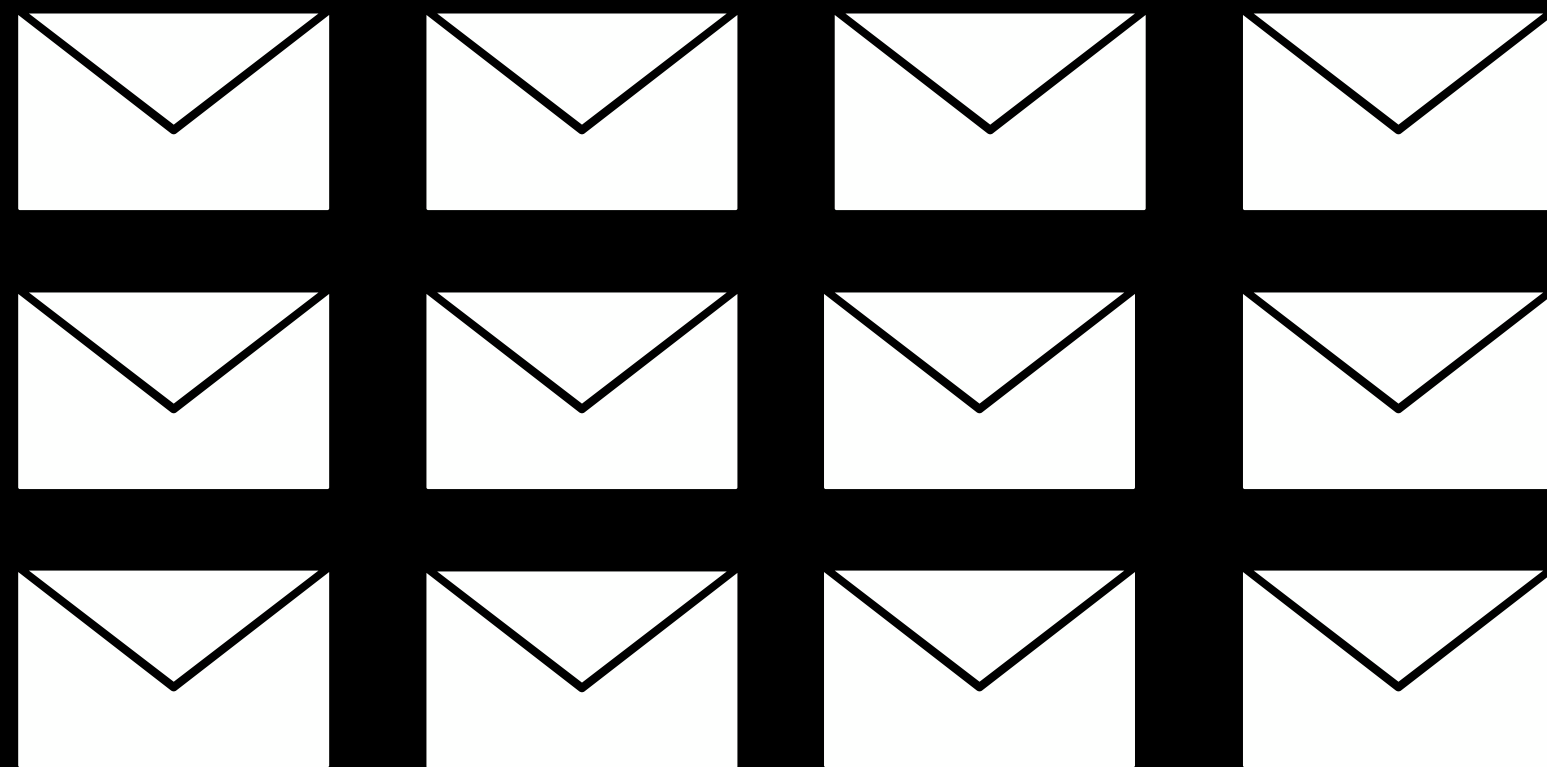
tokenization

the task of splitting a sequence of characters into pieces (tokens)

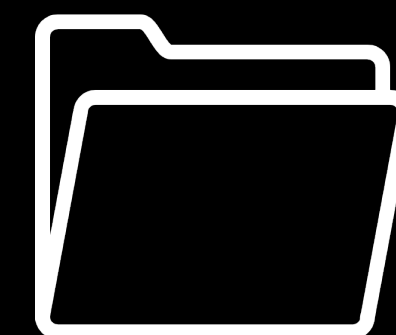
Markov Chains



Text Categorization



Inbox



Spam



"My grandson loved it! So much fun!"

"Product broke after a few days."

"One of the best games I've played in a long time."

"Kind of cheap and flimsy, not worth it."



"My grandson loved it! So much fun!"



"Product broke after a few days."



"One of the best games I've played in a long time."



"Kind of cheap and flimsy, not worth it."



"My grandson **loved** it! So much **fun**!"



"Product **broke** after a few days."



"One of the **best** games I've played in a long time."



"Kind of **cheap** and **flimsy**, not worth it."

bag-of-words model

model that represents text as an unordered collection of words

Naive Bayes

Bayes' Rule

$$P(b \mid a) = \frac{P(a \mid b) P(b)}{P(a)}$$

$P(\text{Positive})$

$P(\text{Negative})$

$$P(\text{😊})$$

$$P(\text{😞})$$

"My grandson loved it!"

$$P(\text{😊})$$

$P(\text{😊} \mid \text{"my grandson loved it"})$

$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$

$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$

$$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

equal to

$$\frac{P(\text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"} \mid \text{😊})P(\text{😊})}{P(\text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})}$$

$$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

proportional to

$$P(\text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"} \mid \text{😊})P(\text{😊})$$

$$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

proportional to

$$P(\text{😊}, \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

$$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

naively proportional to

$$P(\text{😊})P(\text{"my"} \mid \text{😊})P(\text{"grandson"} \mid \text{😊}) \\ P(\text{"loved"} \mid \text{😊}) P(\text{"it"} \mid \text{😊})$$

$$P(\text{😊}) = \frac{\text{number of positive samples}}{\text{number of total samples}}$$

$$P(\text{"loved"} \mid \text{😊}) = \frac{\text{number of positive samples with "loved"}}{\text{number of positive samples}}$$

$$P(\text{😊})P(\text{"my"} \mid \text{😊})P(\text{"grandson"} \mid \text{😊}) \\ P(\text{"loved"} \mid \text{😊}) P(\text{"it"} \mid \text{😊})$$

😊	😐
0.49	0.51

	😊	😐
my	0.30	0.20
grandson	0.01	0.02
loved	0.32	0.08
it	0.30	0.40

$$P(\text{😊})P(\text{"my"} \mid \text{😊})P(\text{"grandson"} \mid \text{😊}) \\ P(\text{"loved"} \mid \text{😊}) P(\text{"it"} \mid \text{😊})$$

😊	😞
0.49	0.51

	😊	😞
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😊	😞
0.49	0.51

😊 0.00014112

	😊	😞
my	0.30	0.20
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loved	0.32	0.08
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😊 0.00014112

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😄	😞
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😄 0.00014112

	😄	😞
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😄	😞
0.49	0.51

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😄	😞
0.49	0.51

😄 0.00014112

😞 0.00006528

	😄	😞
my	0.30	0.20
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😄	😞
0.49	0.51

😄 0.00014112

😞 0.00006528

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my	0.30	0.20
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😄	😞
0.49	0.51

😄 0.6837

😞 0.3163

	😄	😞
my	0.30	0.20
grandson	0.01	0.02
loved	0.32	0.08
it	0.30	0.40

$$P(\text{😞})P(\text{"my"} \mid \text{😞})P(\text{"grandson"} \mid \text{😞}) \\ P(\text{"loved"} \mid \text{😞}) P(\text{"it"} \mid \text{😞})$$

😄	😞
0.49	0.51

	😄	😞
my	0.30	0.20
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😄	😞
0.49	0.51

😄 0.000000000

😞 0.00006528

	😄	😞
my	0.30	0.20
grandson	0.00	0.02
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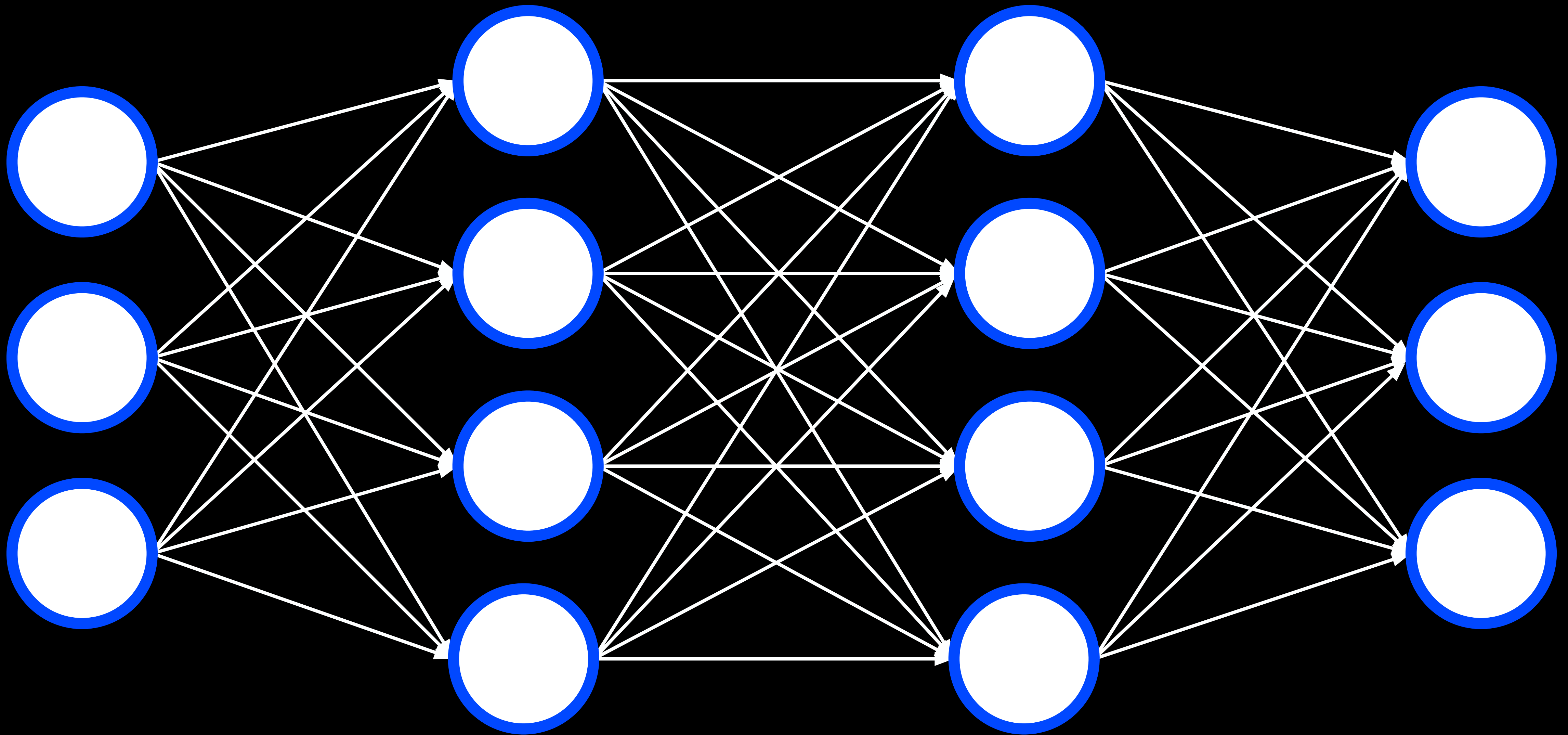
additive smoothing

adding a value α to each value in our distribution to smooth the data

Laplace smoothing

adding 1 to each value in our distribution:
pretending we've seen each value one more
time than we actually have

Word Representation



"He wrote a book."

he	[1, 0, 0, 0]
----	--------------

wrote	[0, 1, 0, 0]
-------	--------------

a	[0, 0, 1, 0]
---	--------------

book	[0, 0, 0, 1]
------	--------------

one-hot representation

representation of meaning as a vector with a single 1, and with other values as 0

"He wrote a book."

he	[1, 0, 0, 0]
----	--------------

wrote	[0, 1, 0, 0]
-------	--------------

a	[0, 0, 1, 0]
---	--------------

book	[0, 0, 0, 1]
------	--------------

"He wrote a book."

he	[1, 0, 0, 0, 0, 0, 0, 0, ...]
wrote	[0, 1, 0, 0, 0, 0, 0, 0, ...]
a	[0, 0, 1, 0, 0, 0, 0, 0, ...]
book	[0, 0, 0, 1, 0, 0, 0, 0, ...]

"He wrote a book."

"He authored a novel."

wrote	[0, 1, 0, 0, 0, 0, 0, ...]
-------	----------------------------

authored	[0, 0, 0, 0, 1, 0, 0, ...]
----------	----------------------------

book	[0, 0, 0, 1, 0, 0, 0, ...]
------	----------------------------

novel	[0, 0, 0, 0, 0, 0, 1, ...]
-------	----------------------------

distributed representation

representation of meaning distributed
across multiple values

"He wrote a book."

he	$[-0.34, -0.08, 0.02, -0.18, 0.22, \dots]$
wrote	$[-0.27, 0.40, 0.00, -0.65, -0.15, \dots]$
a	$[-0.12, -0.25, 0.29, -0.09, 0.40, \dots]$
book	$[-0.23, -0.16, -0.05, -0.57, 0.05, \dots]$

"You shall know a word
by the company it keeps."

J. R. Firth, 1957

for		he	ate
-----	--	----	-----

for

breakfast

he

ate

for

lunch

he

ate

for

dinner

he

ate

for		he	ate
-----	--	----	-----

word2vec

model for generating word vectors

breakfast

book

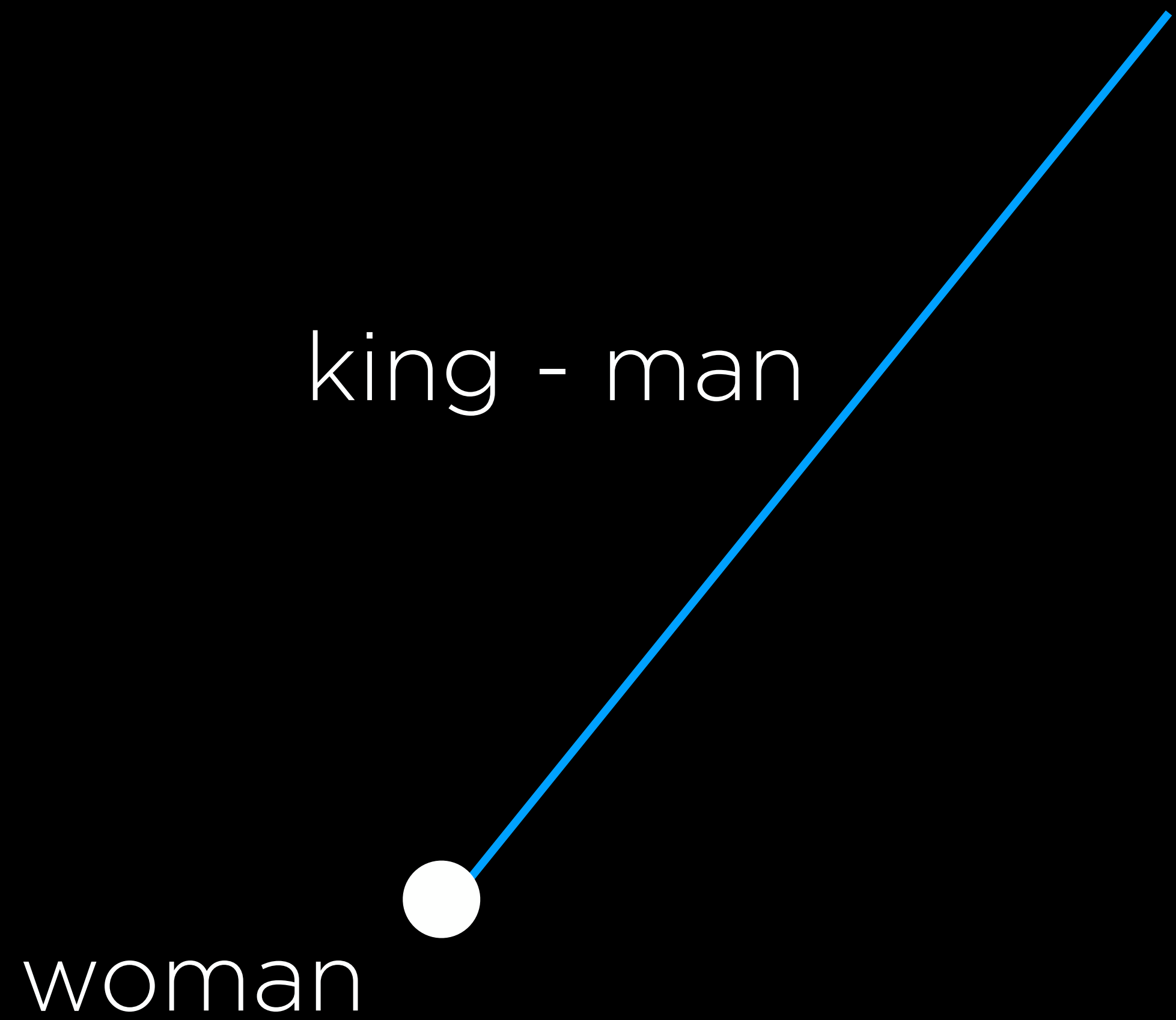
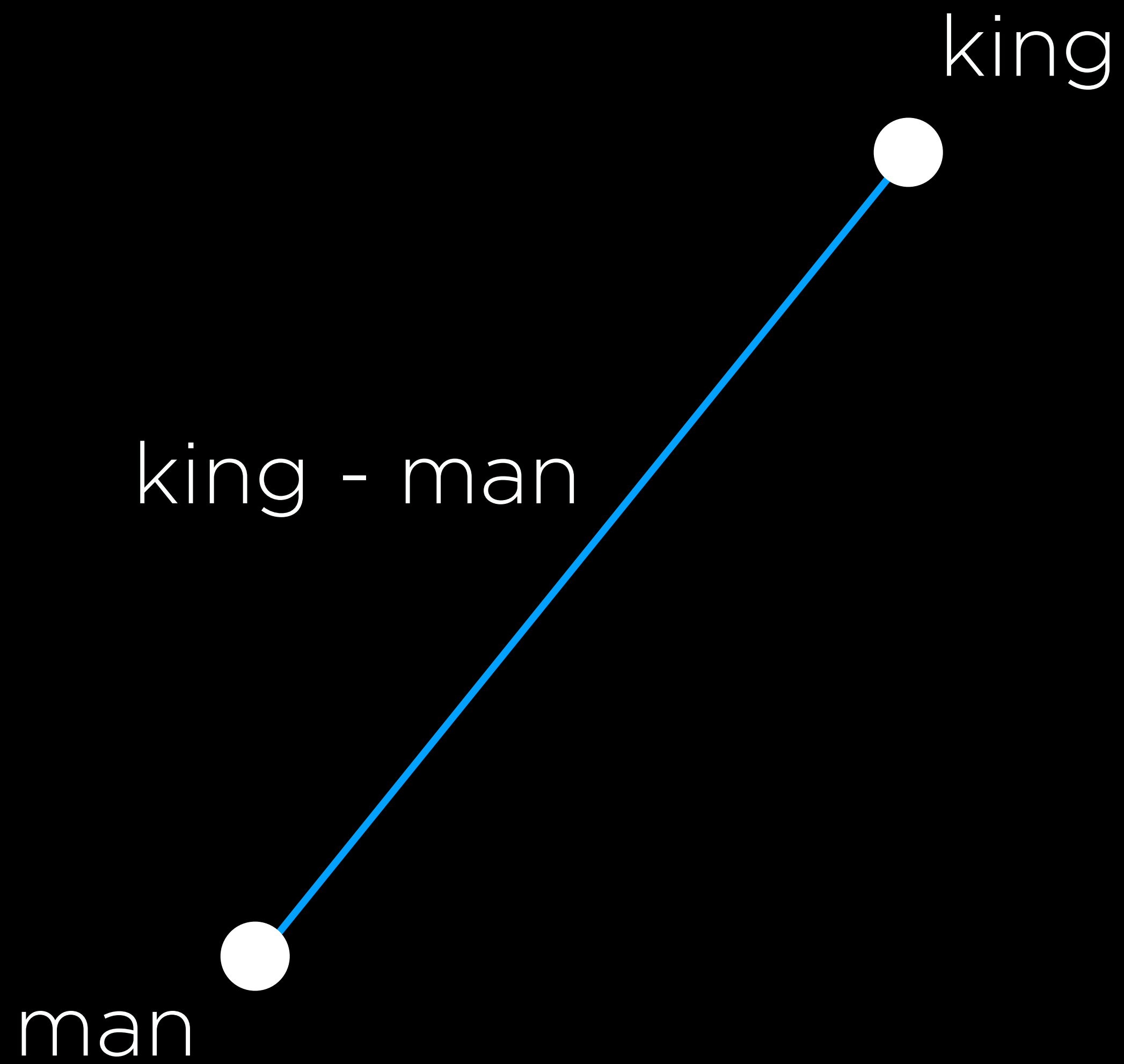
memoir

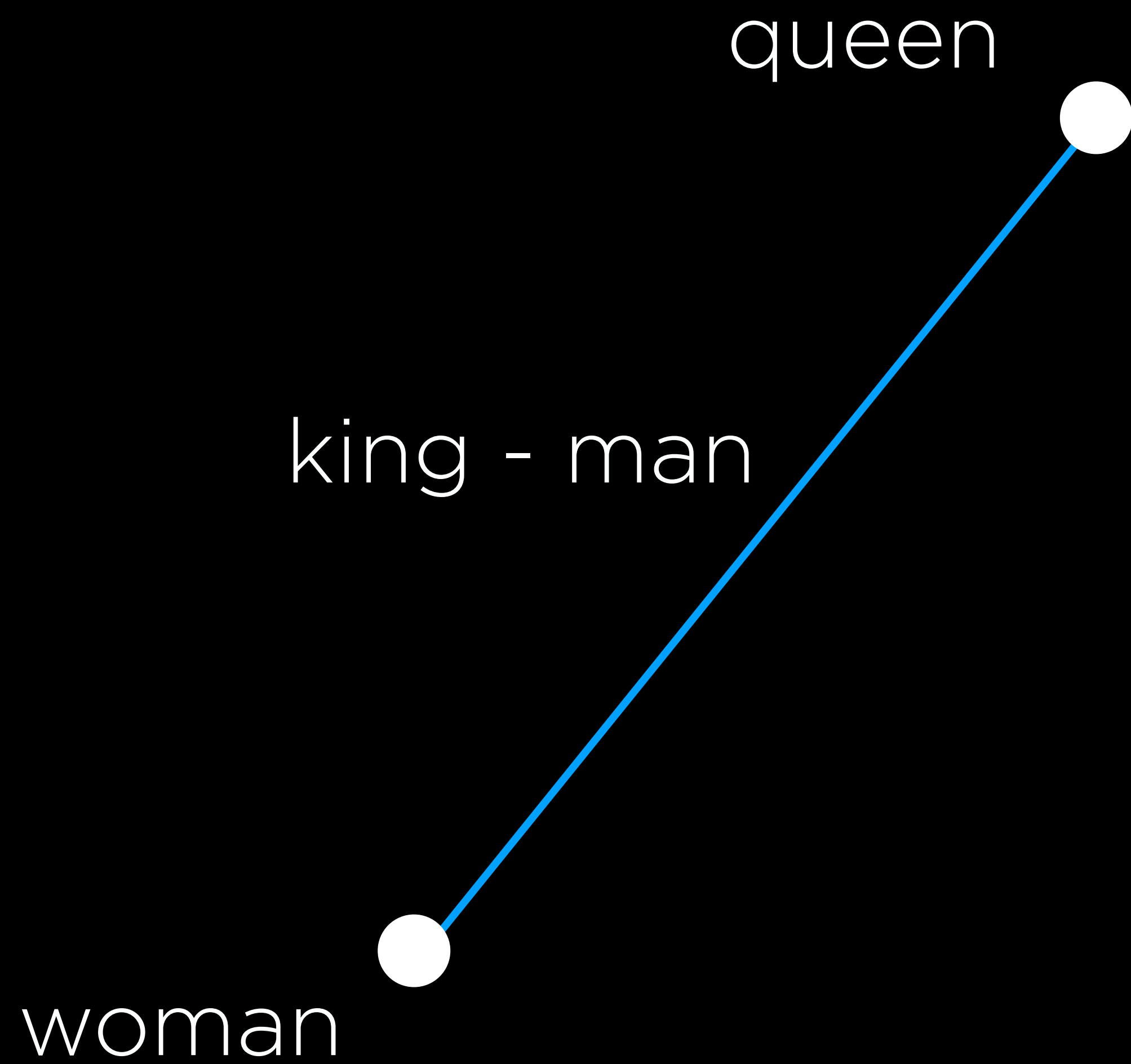
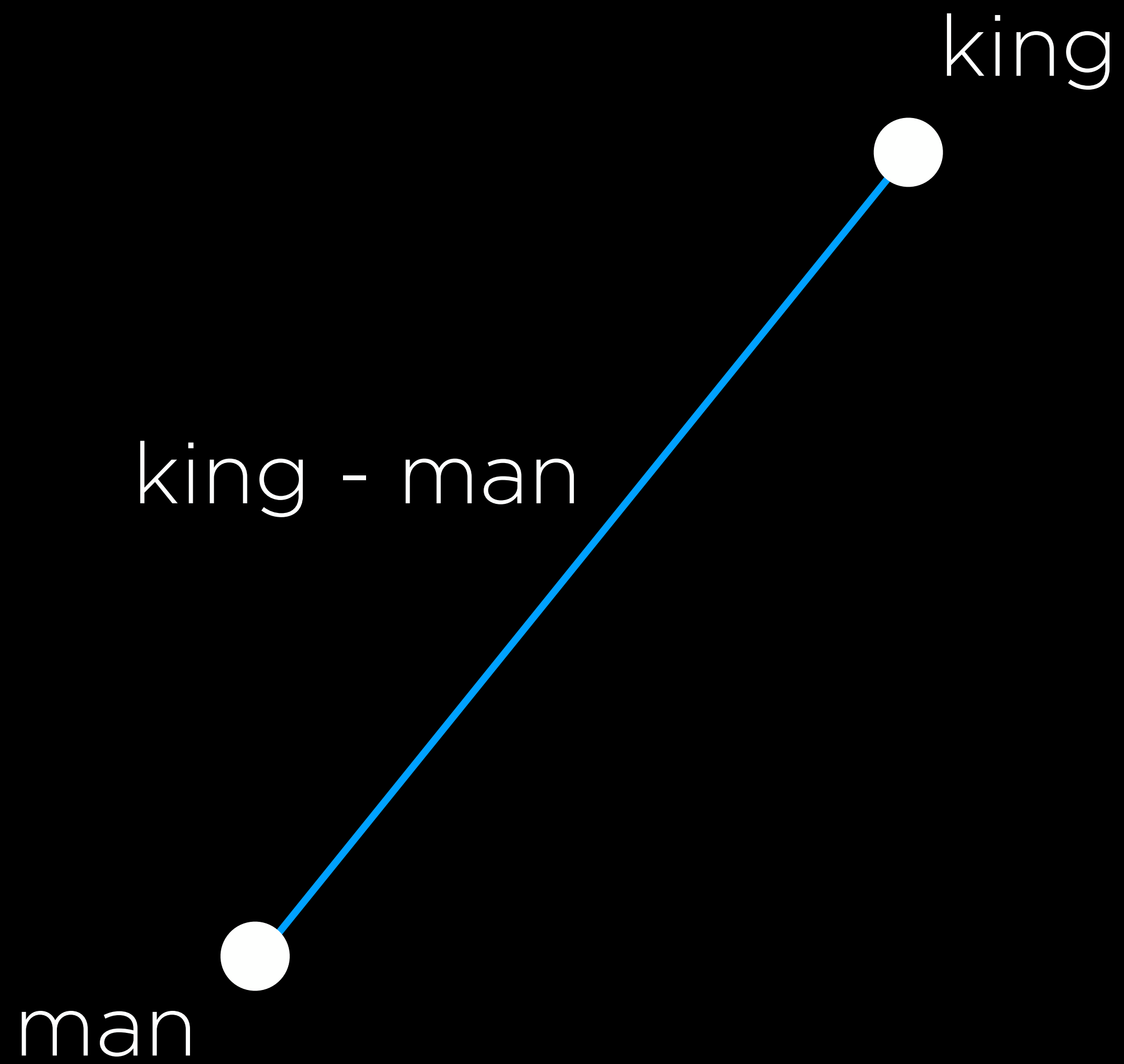
lunch

dinner

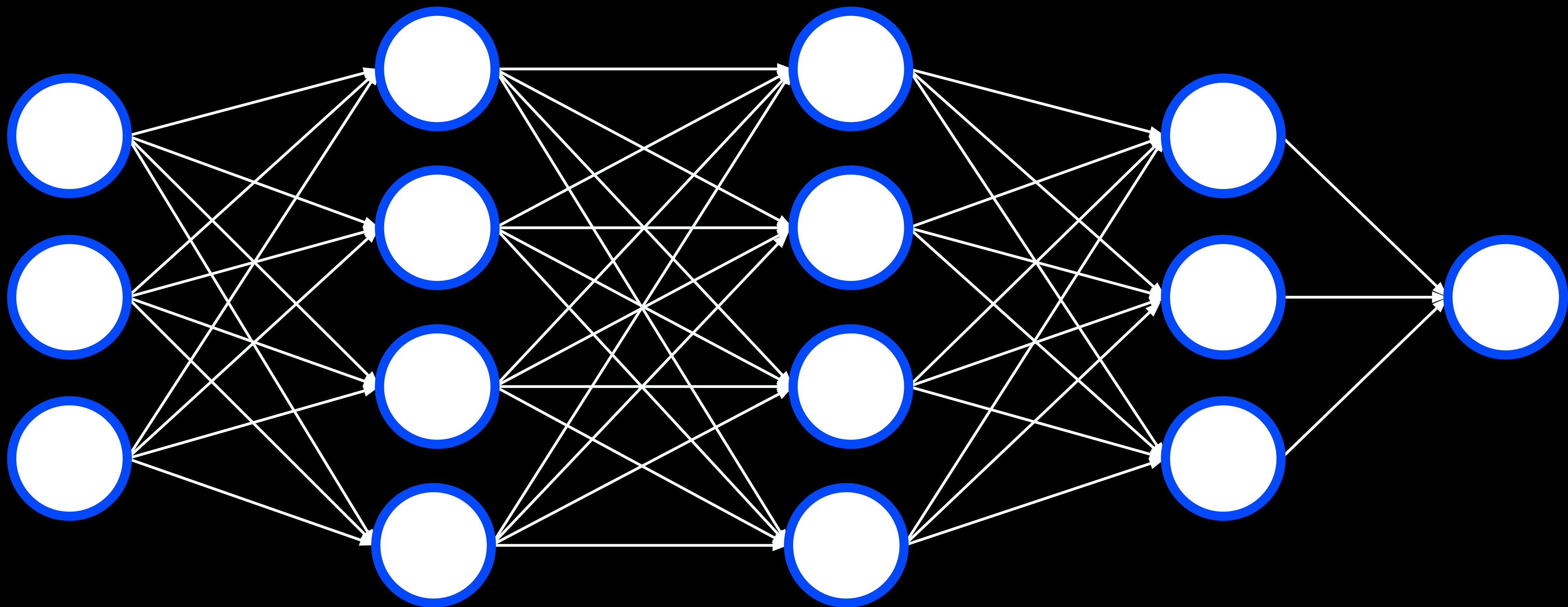
novel

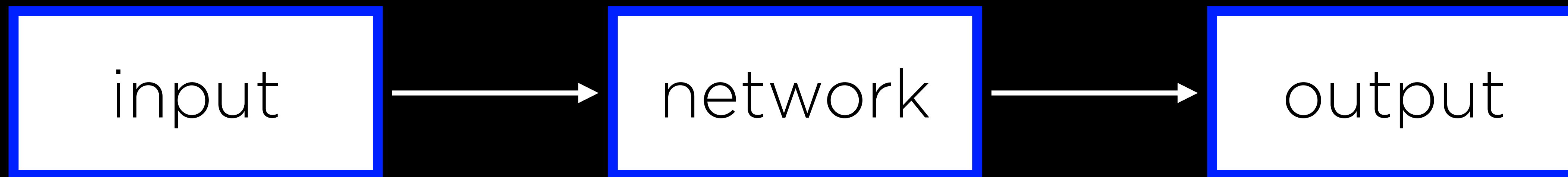


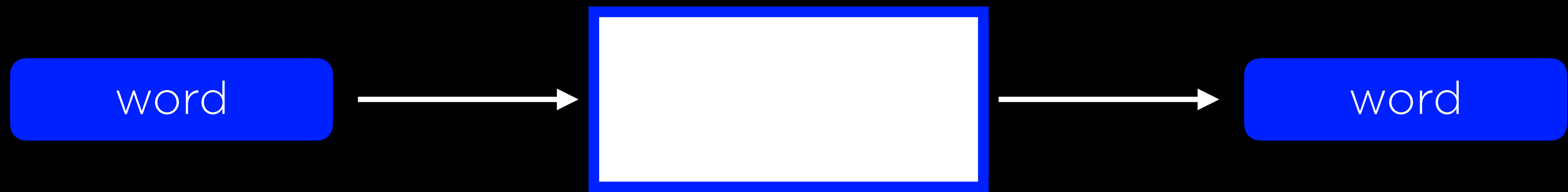




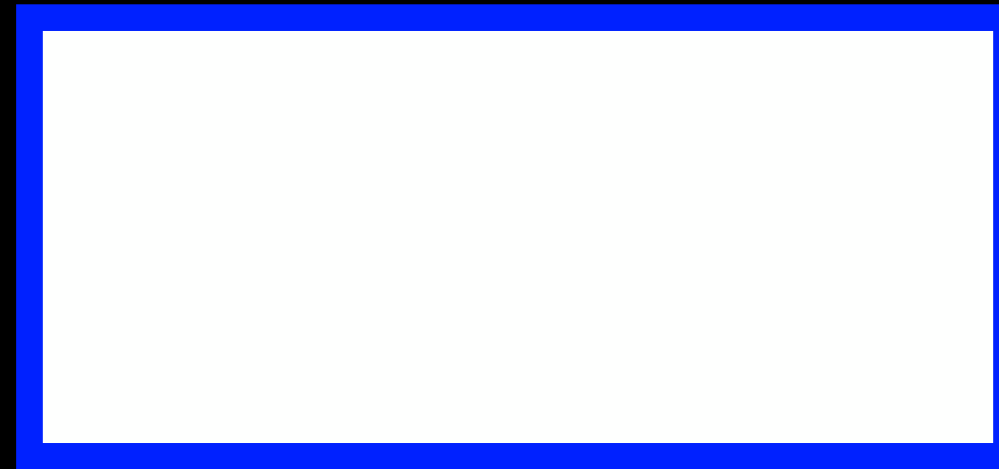
Neural Networks





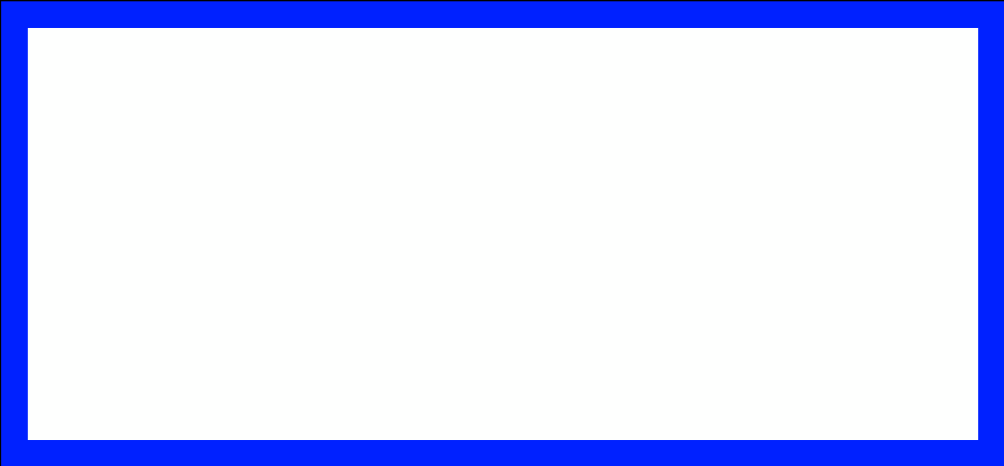


English



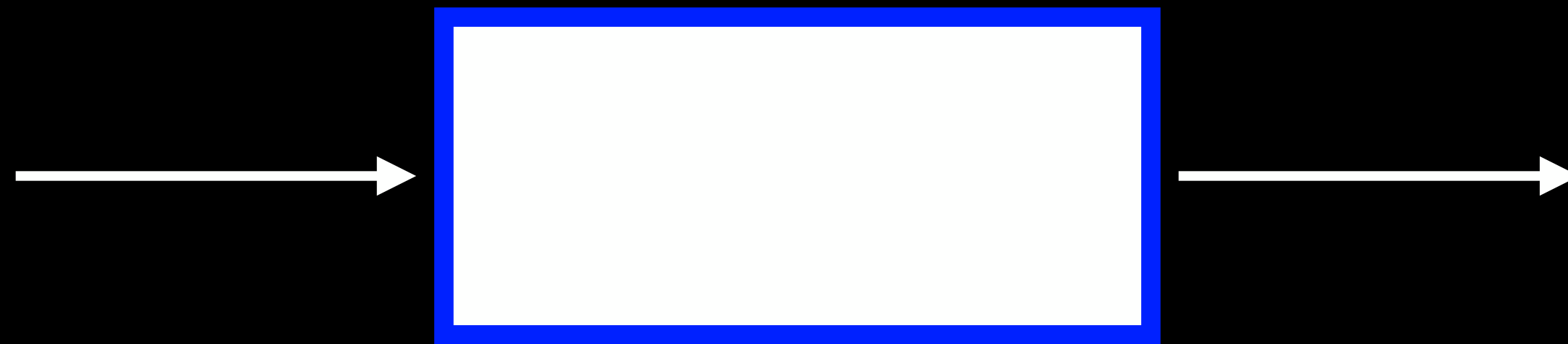
French

lamp



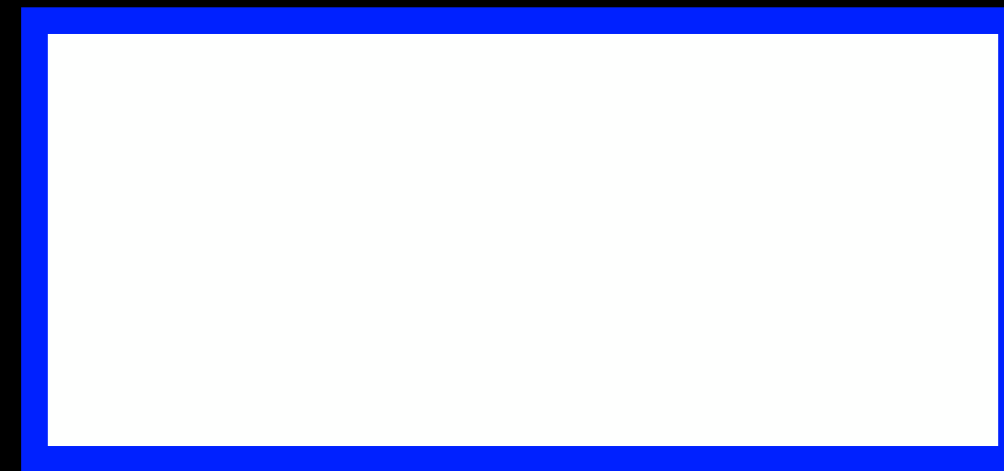
lampe

The only light in
the room came
from the lamp
upon the table at
which I had been
reading.

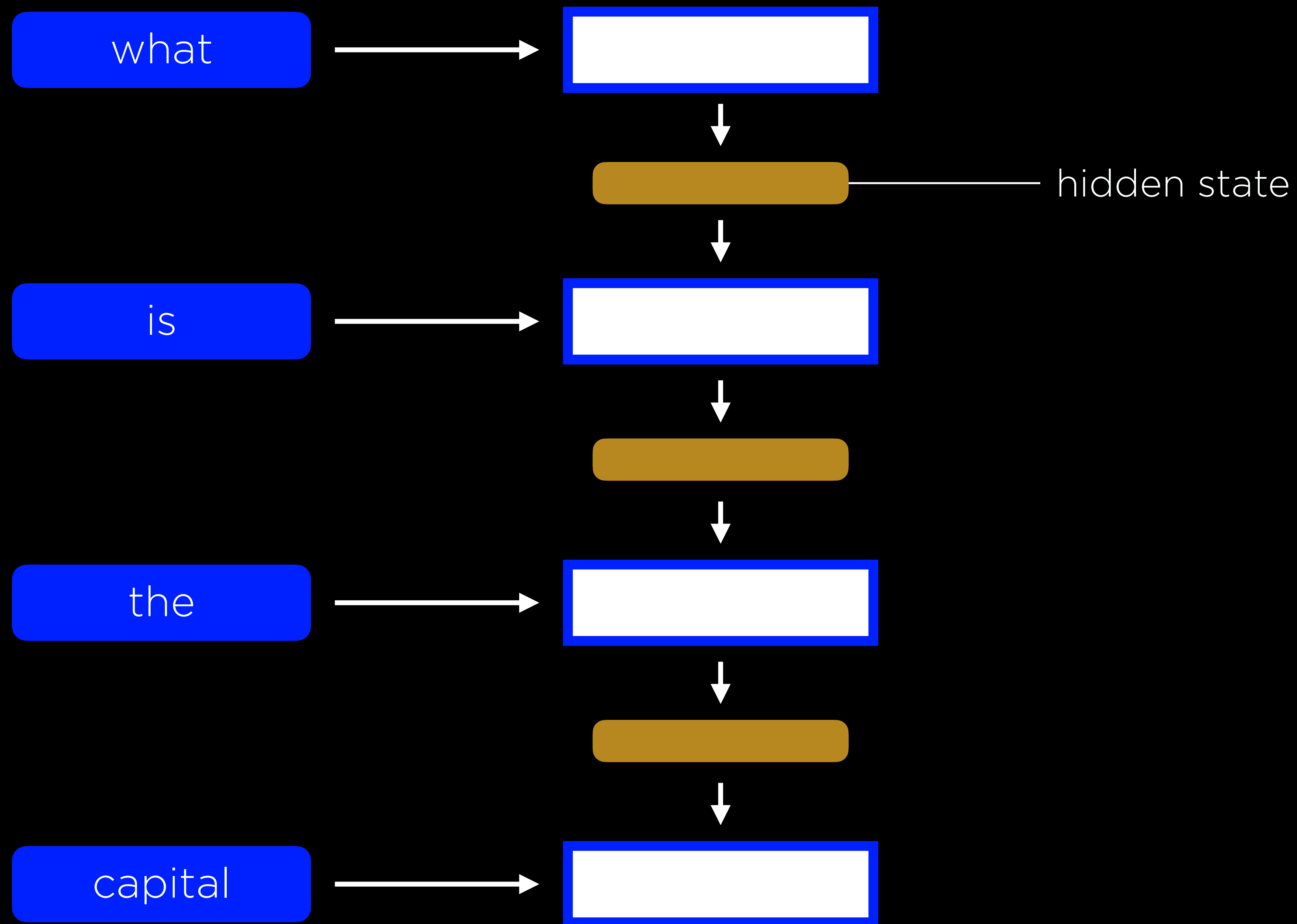


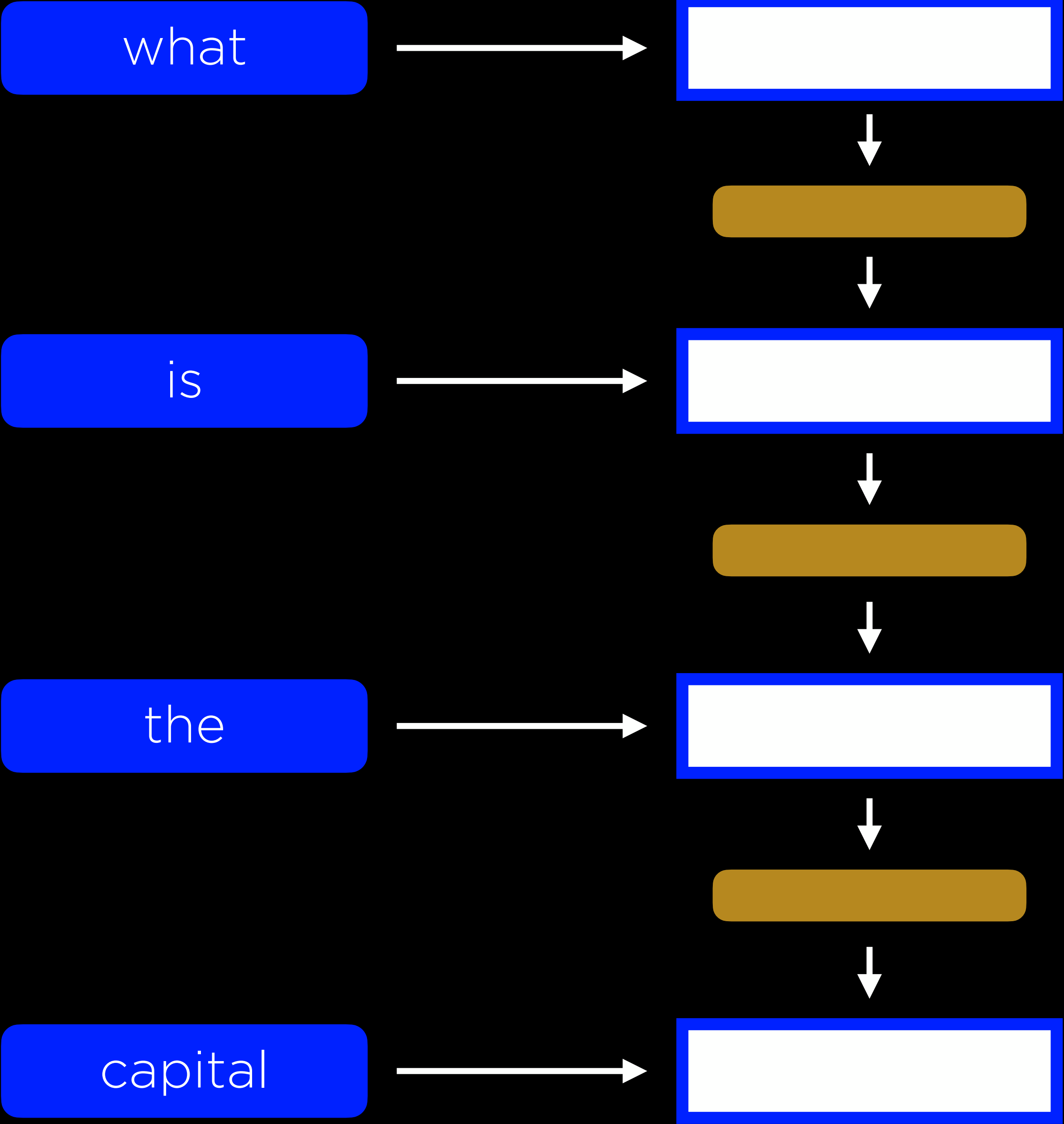
La pièce n'était
éclairée que par
la lampe placée
sur la table où je
lisais.

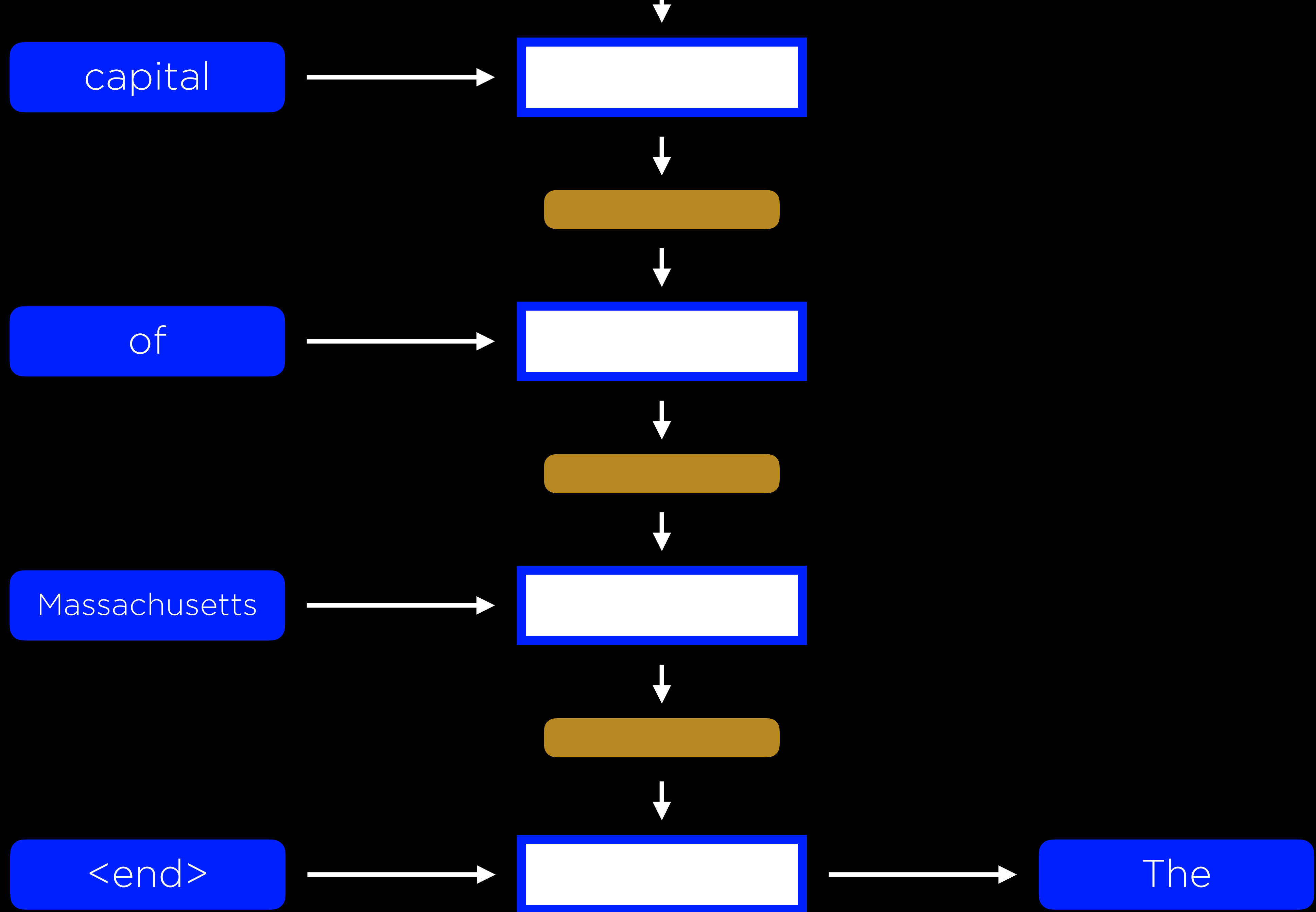
What is the
capital of
Massachusetts?

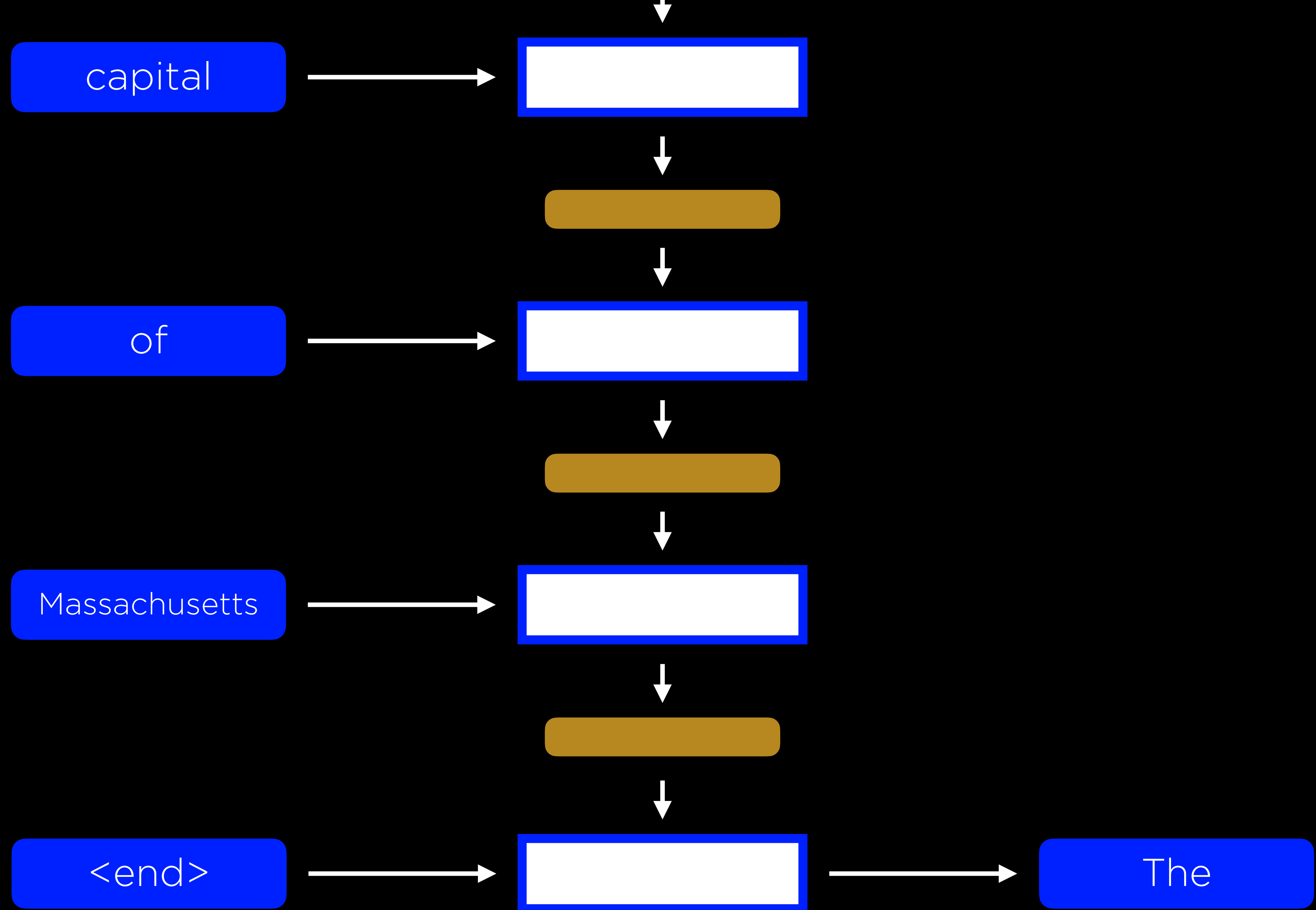


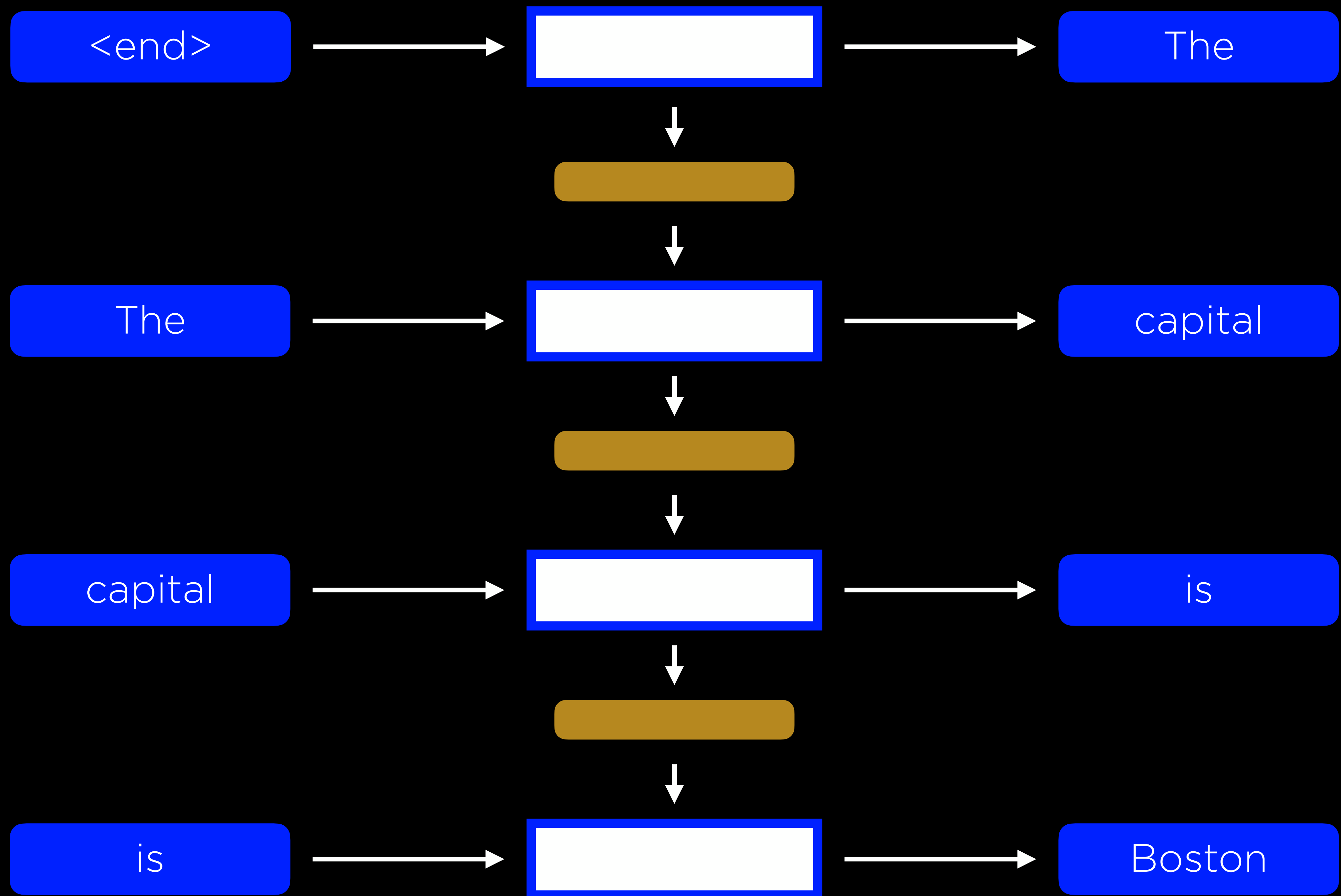
The capital
is Boston.

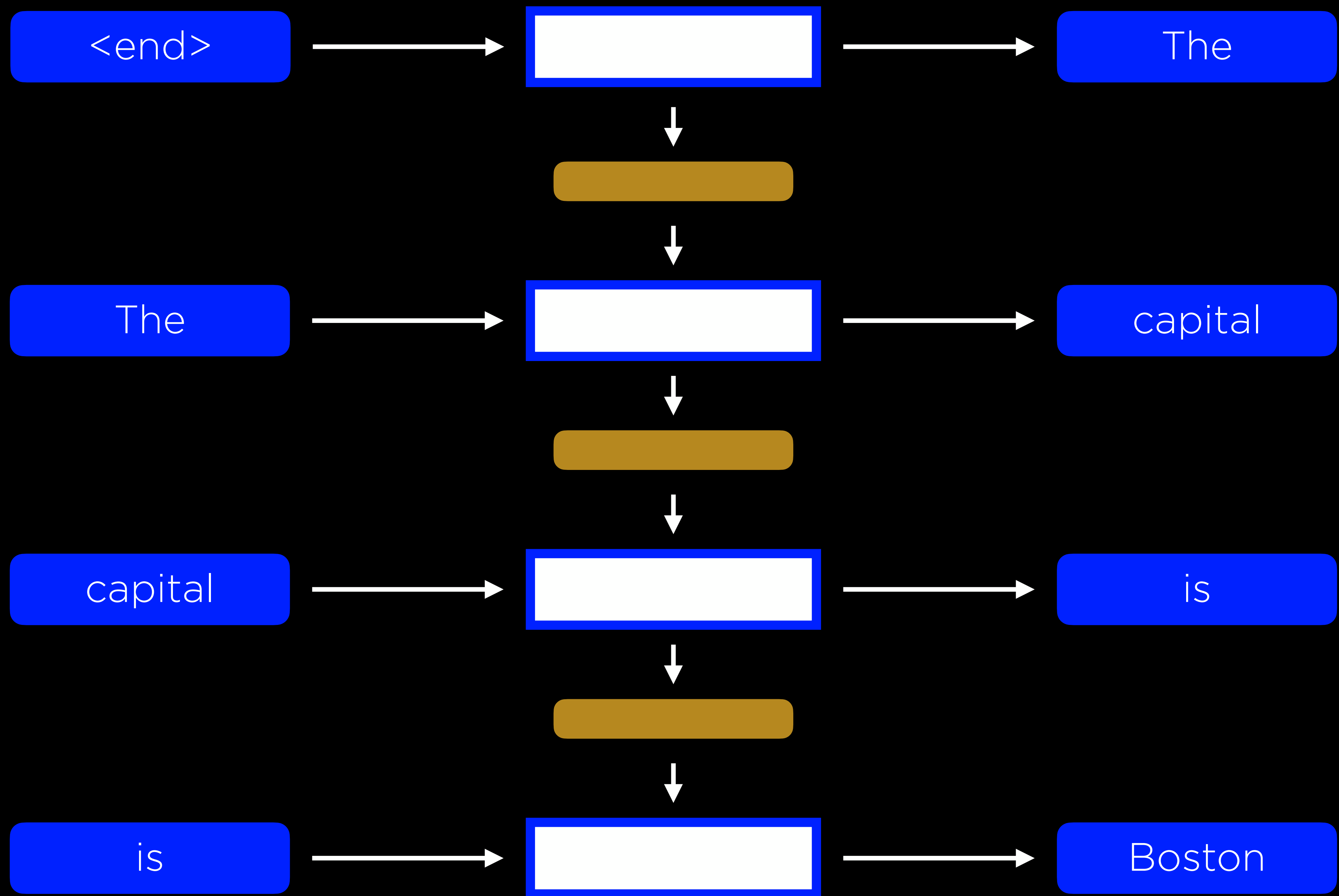


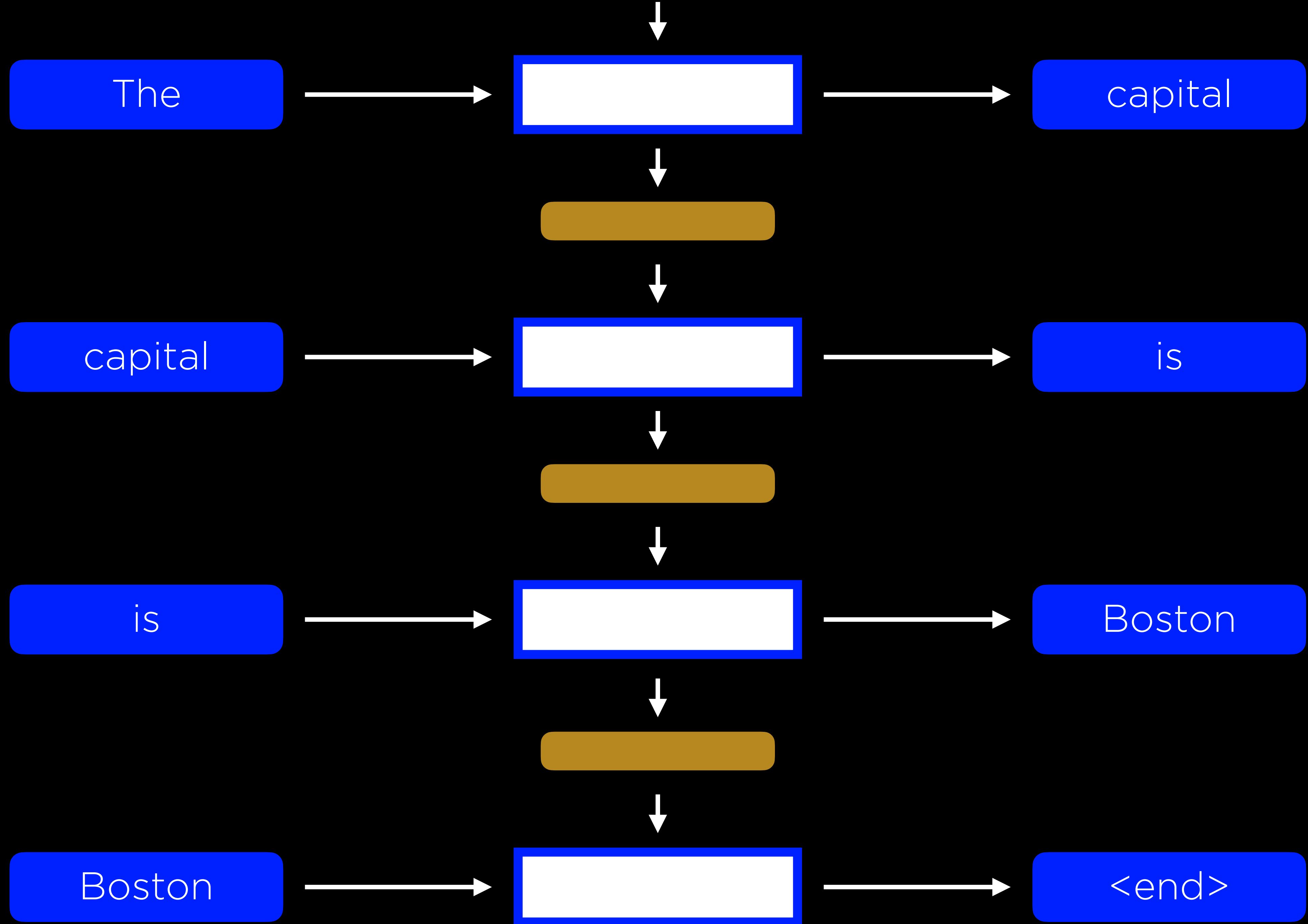


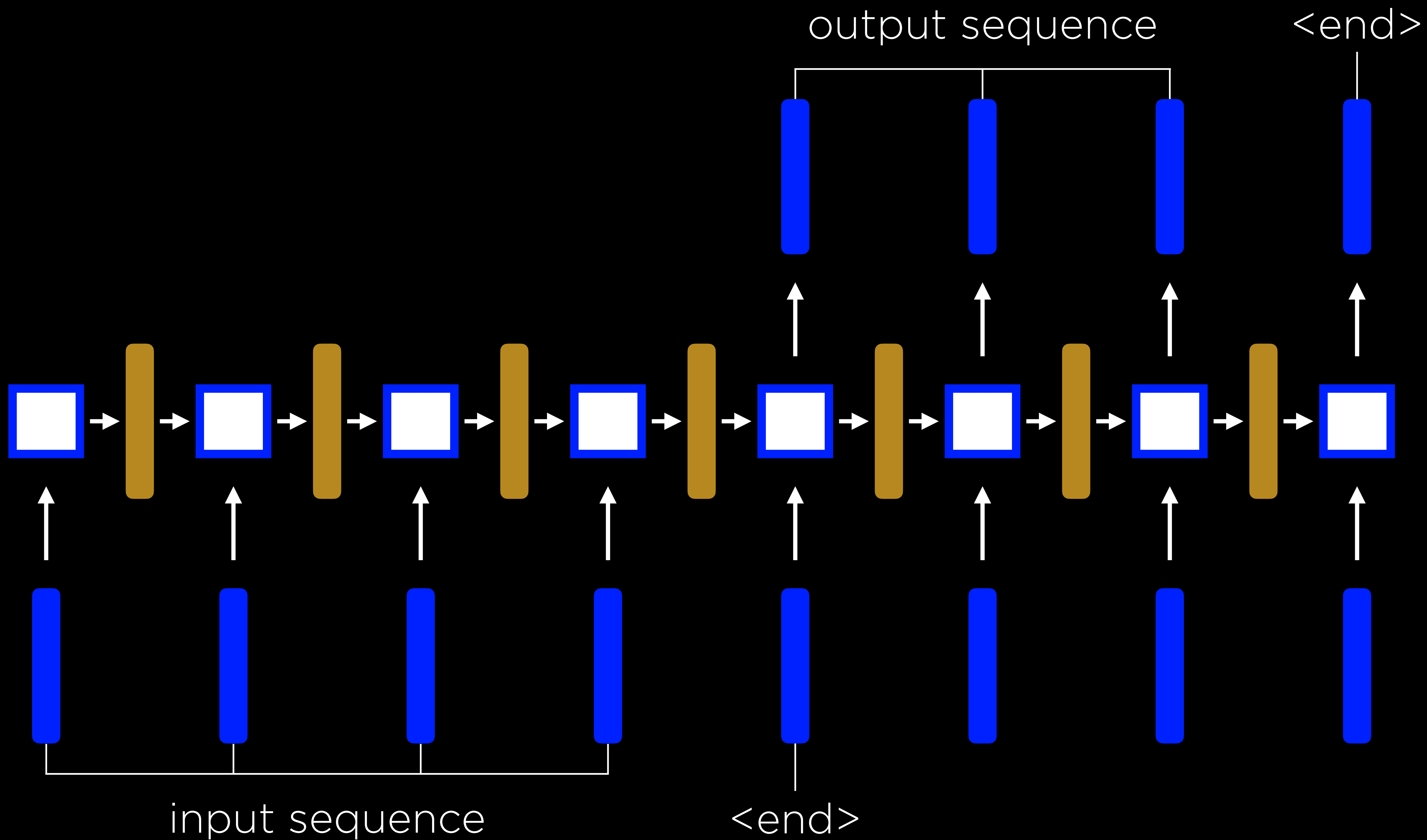


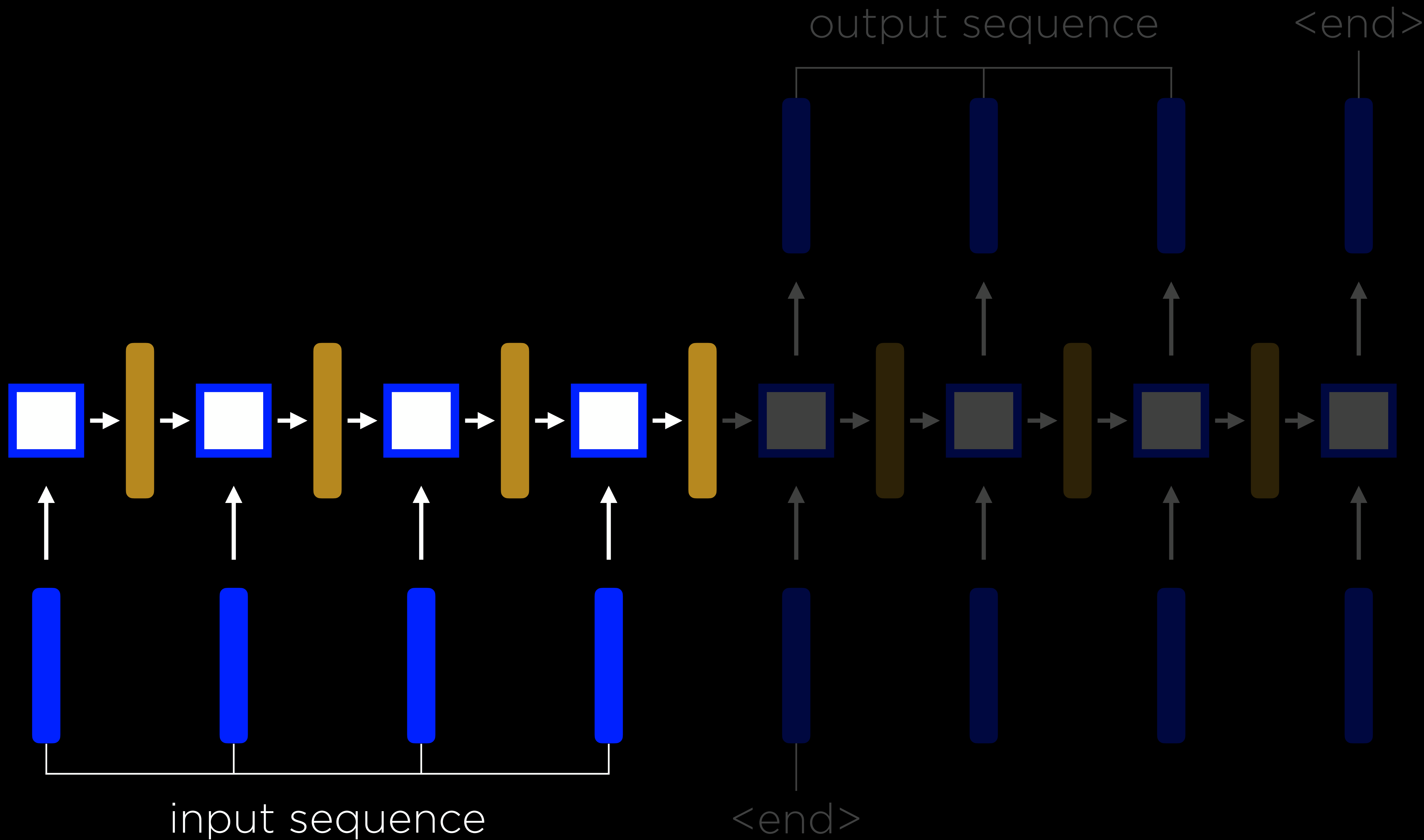


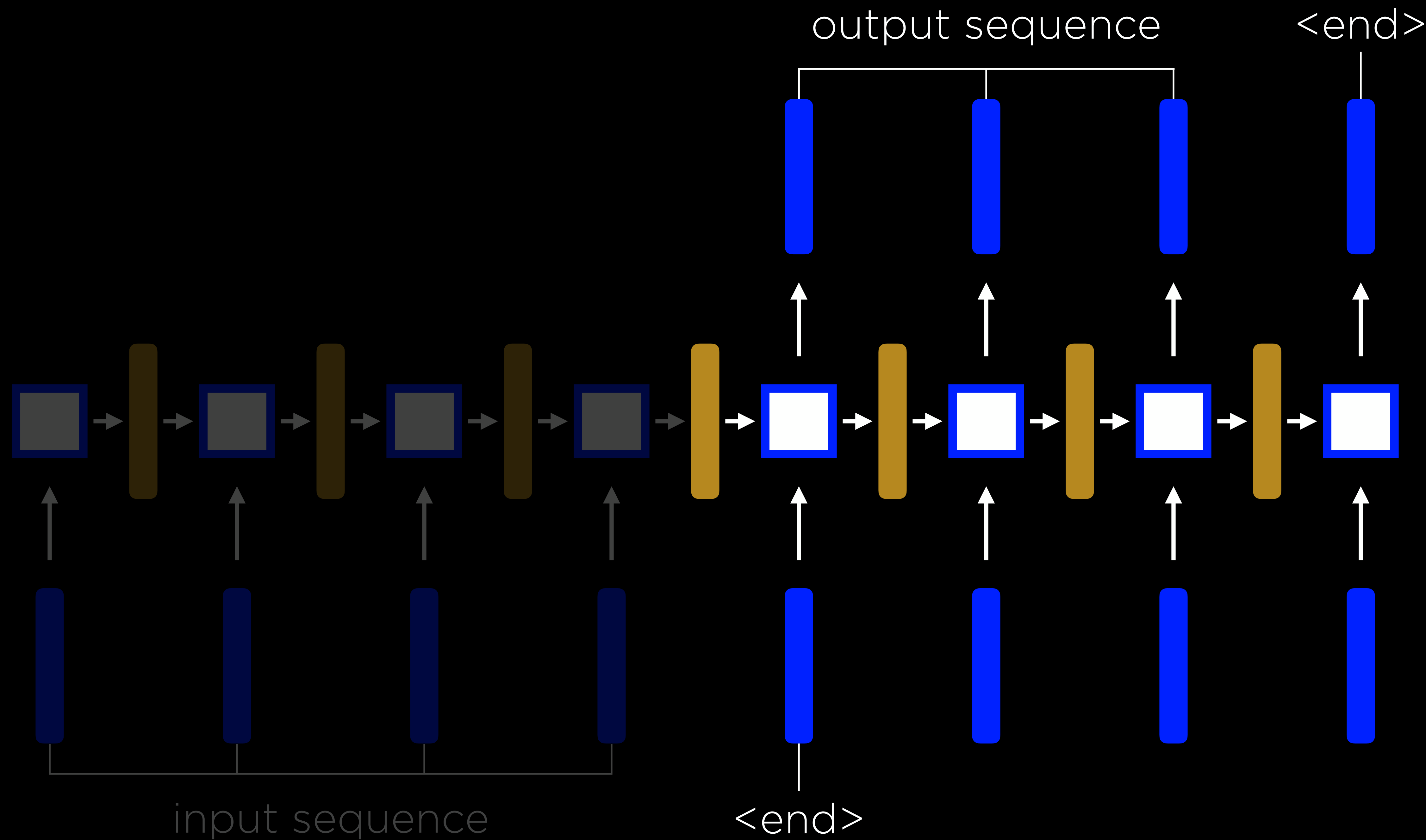


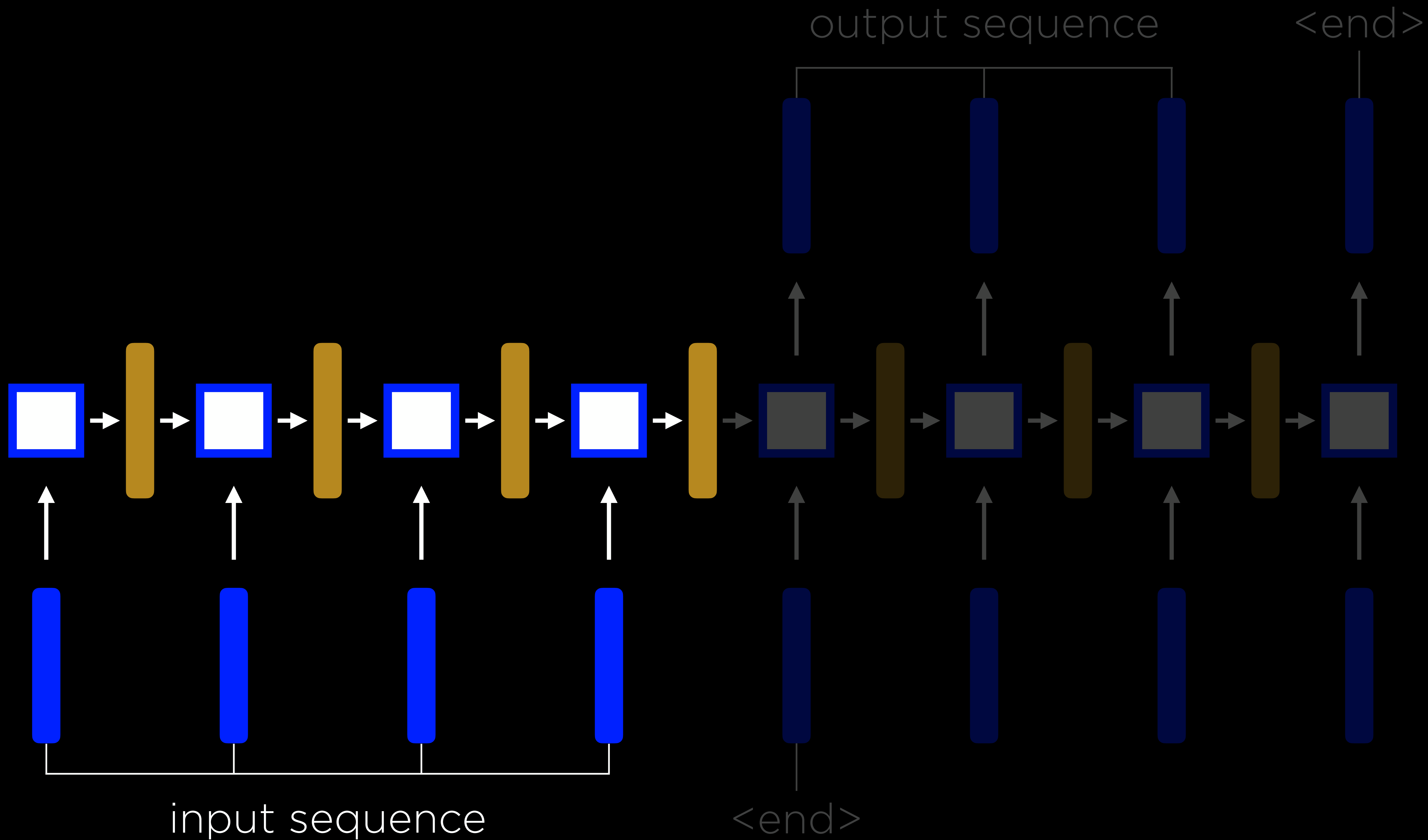


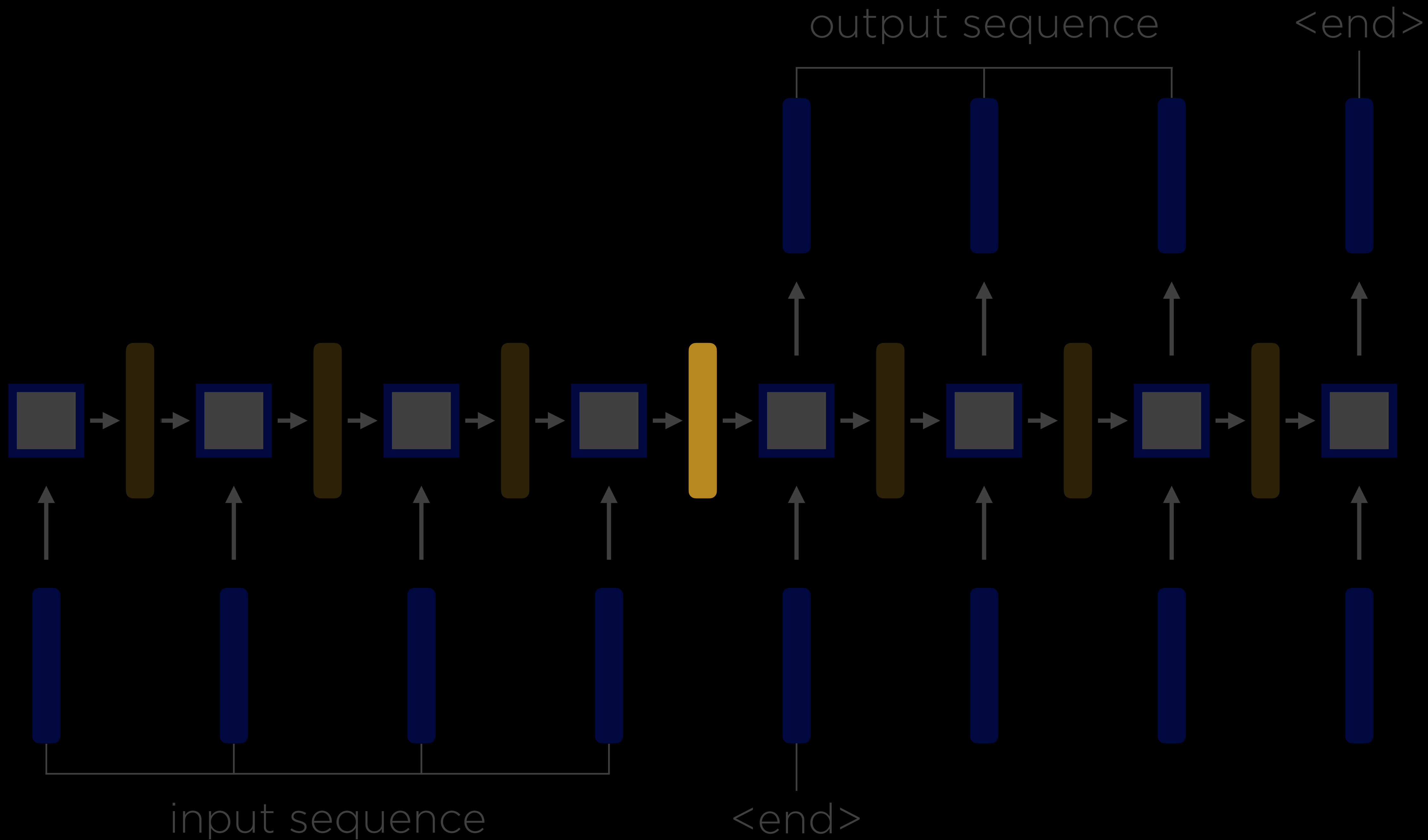


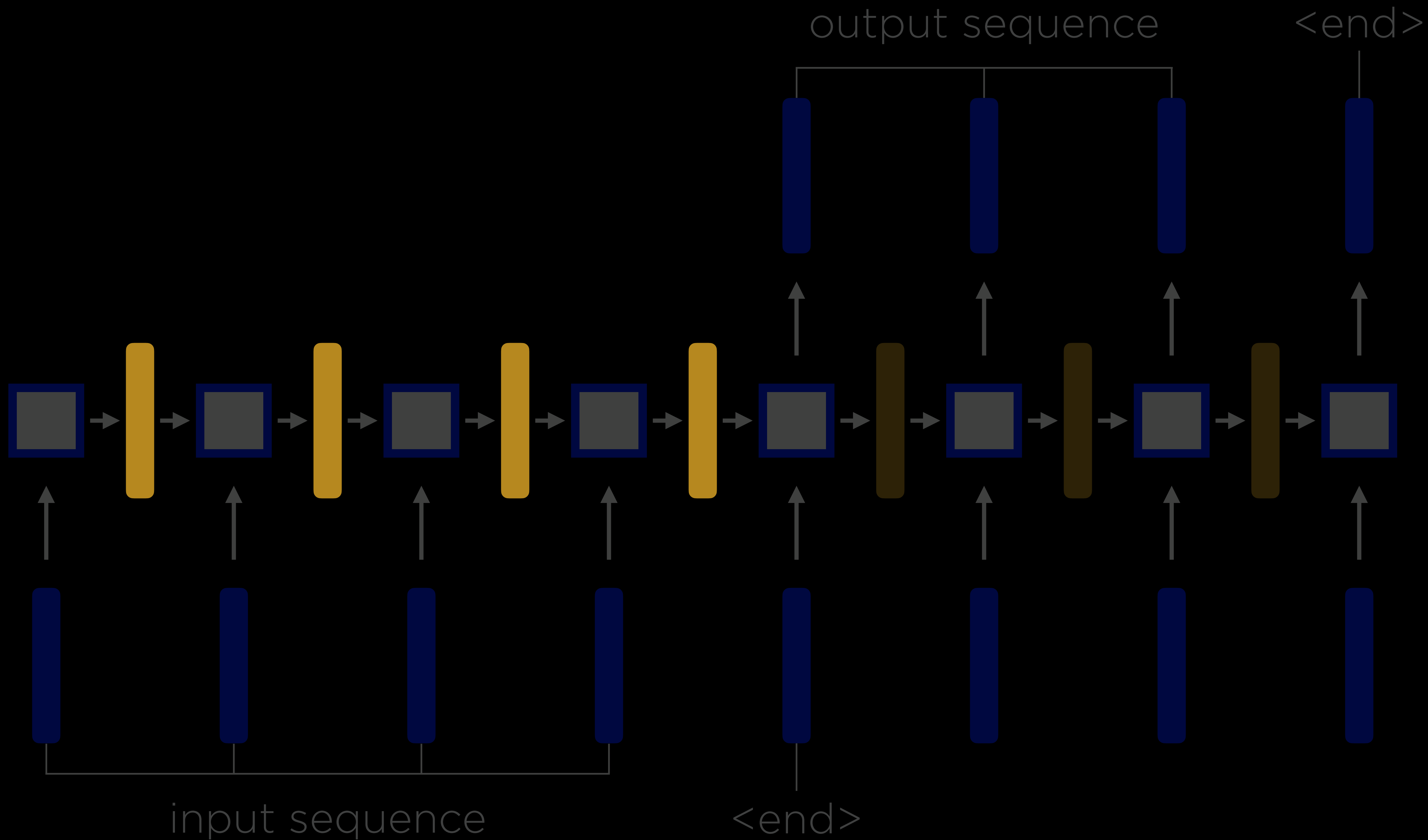












Attention

the

capital

is

what

is

the

capital

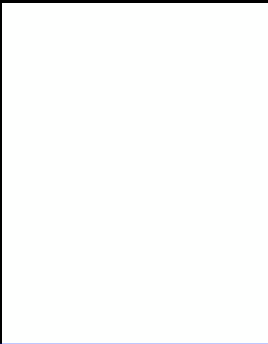
of

Massachusetts

the

capital

is



what



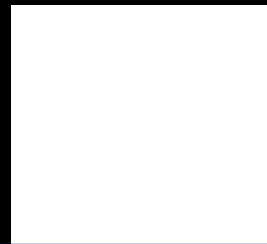
is



the



capital



of



Massachusetts

the

capital

is



+



+



+



+



+



=



x

x

x

x

x

x

0.04

0.02

0.01

0.28

0.03

0.62

what

is

the

capital

of

Massachusetts

the

capital

is

Boston



+



+



+



+



+



=



x

0.04

x

0.02

x

0.01

x

0.28

x

0.03

x

0.62

what

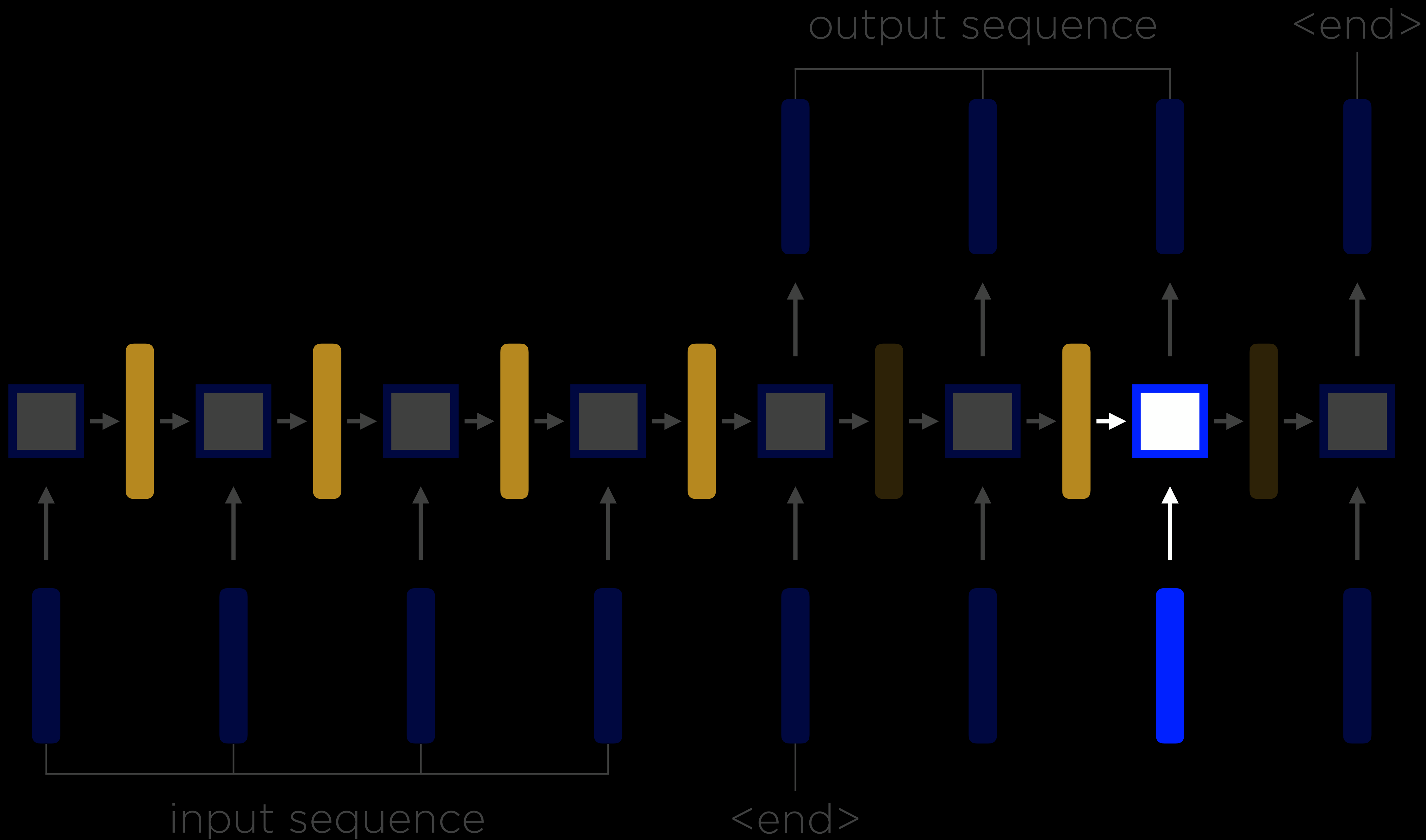
is

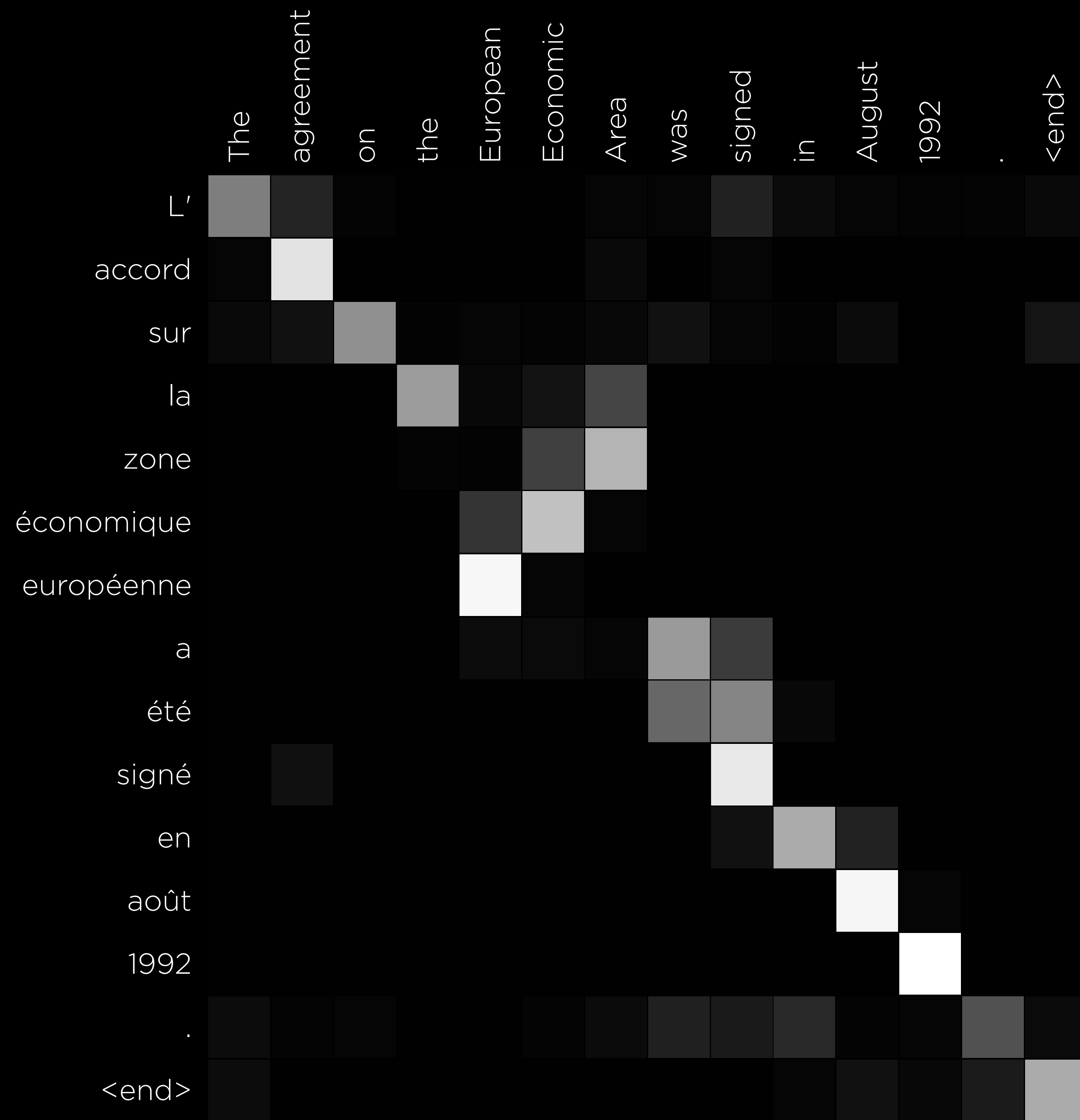
the

capital

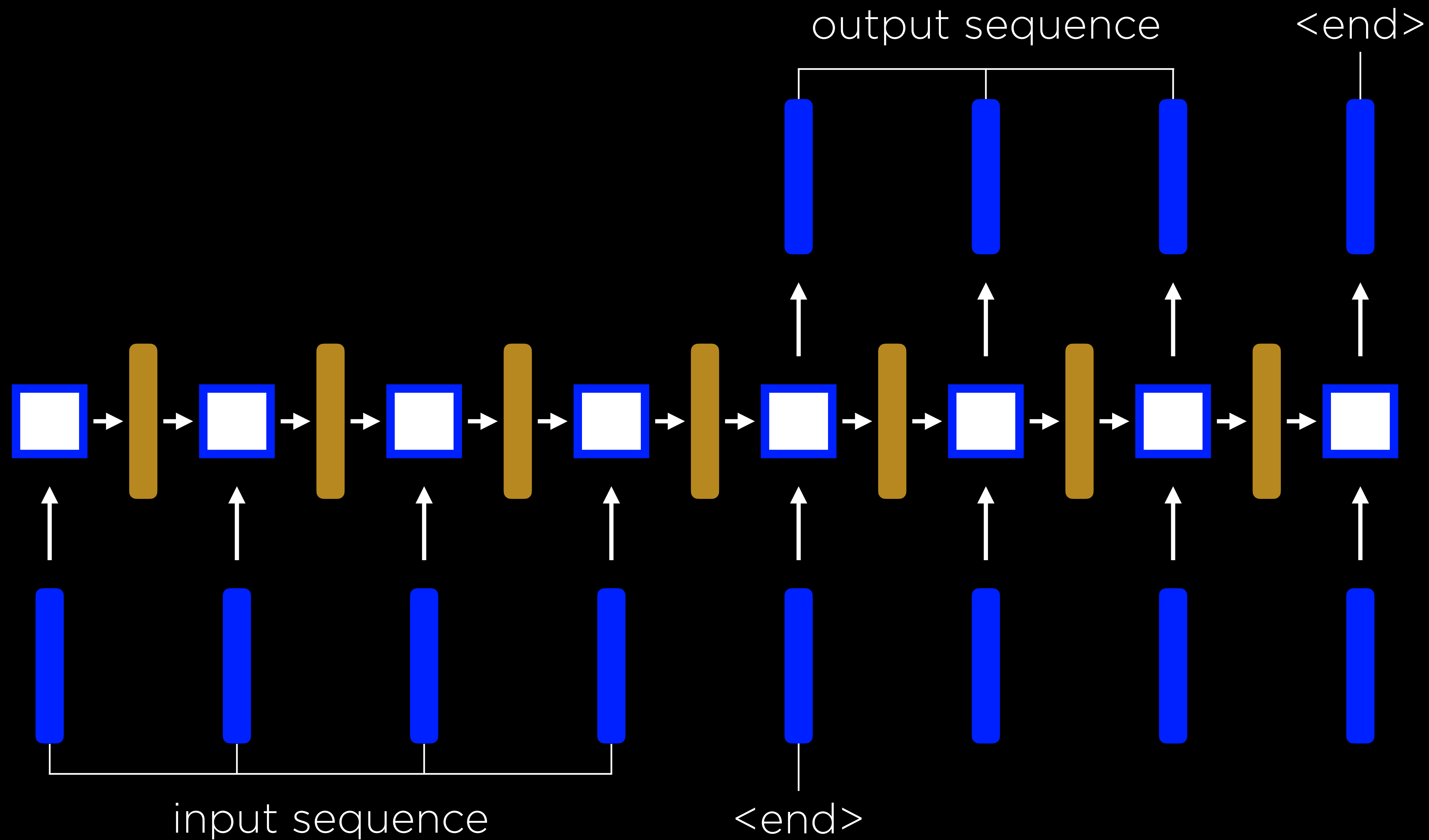
of

Massachusetts

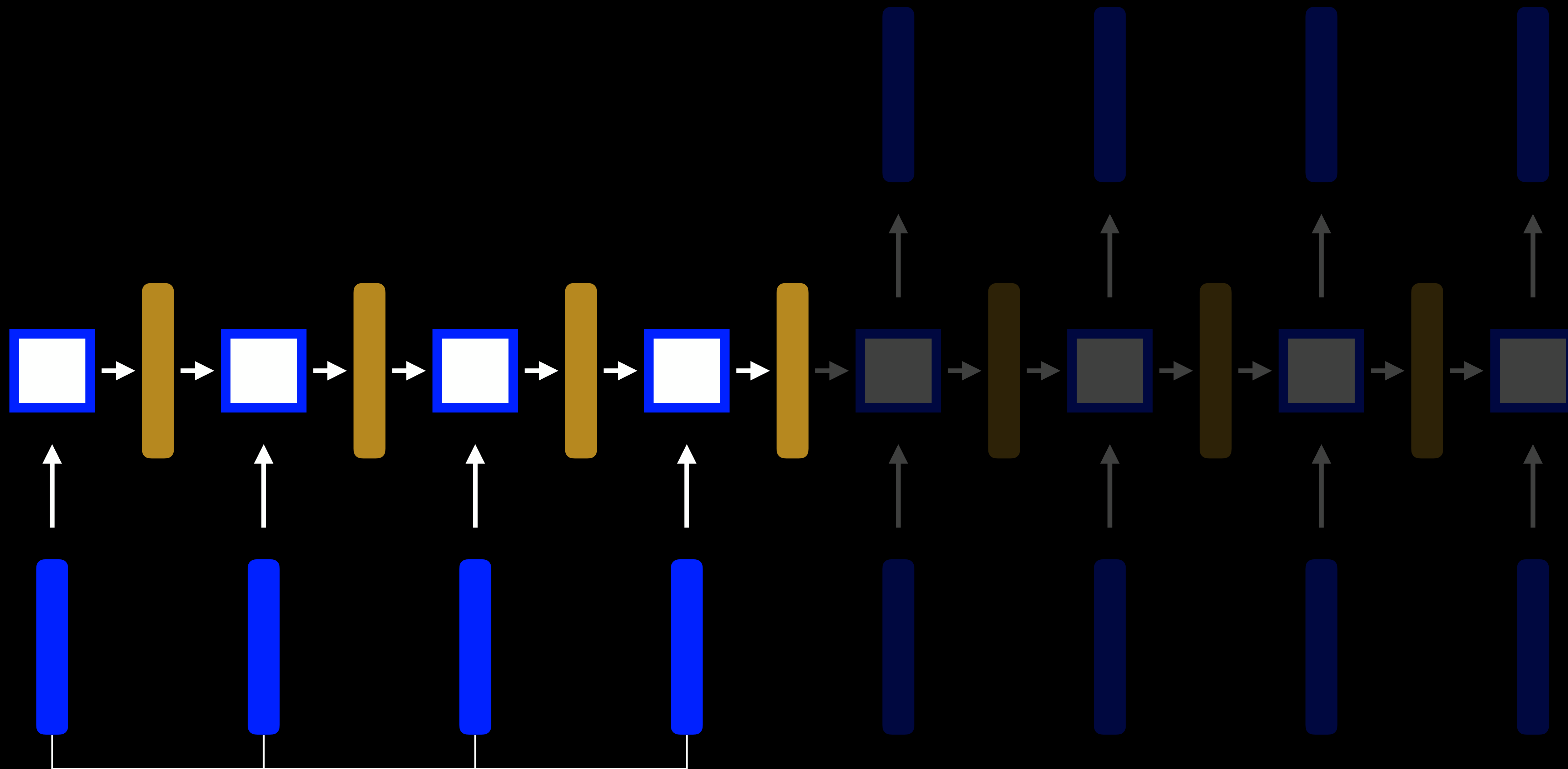




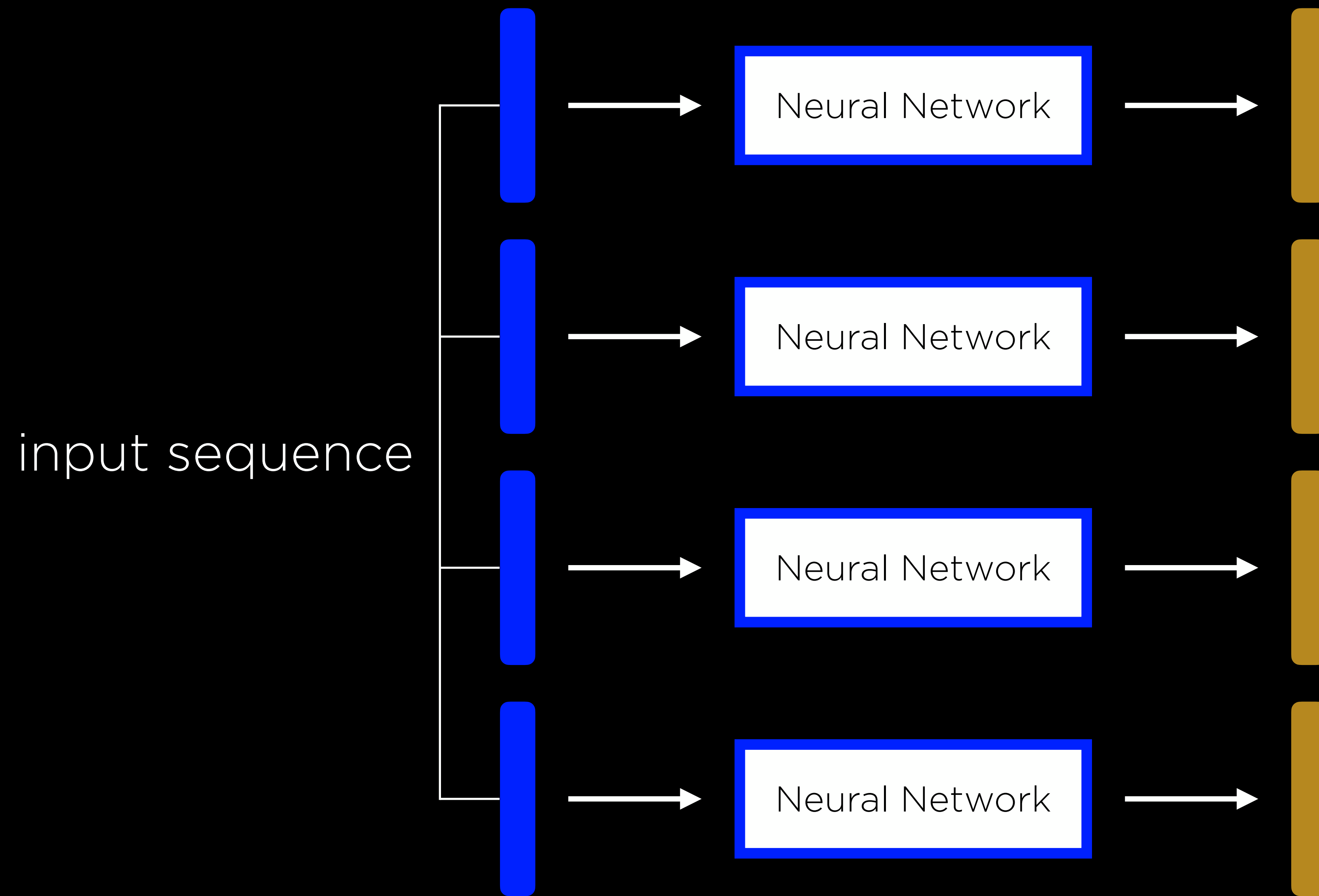
Adapted from Bahdanau et al. 2015.
Neural machine translation by jointly
learning to align and translate

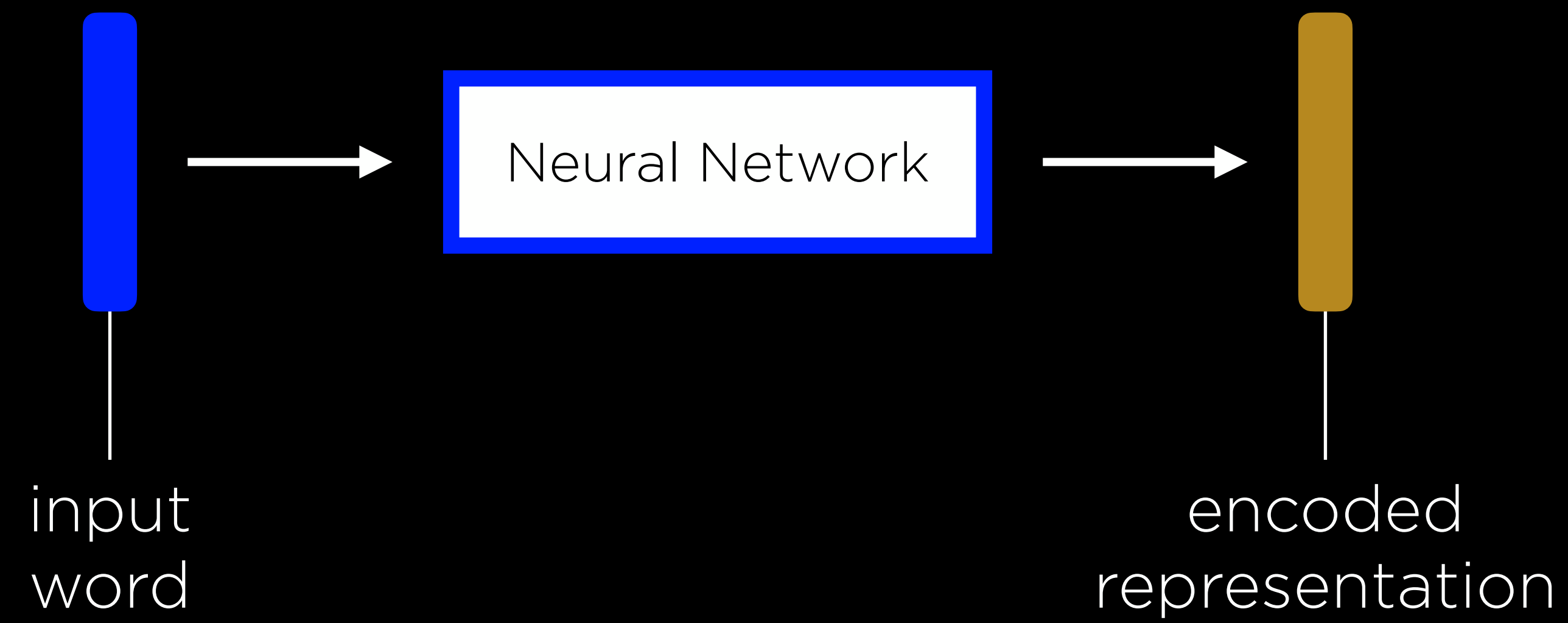


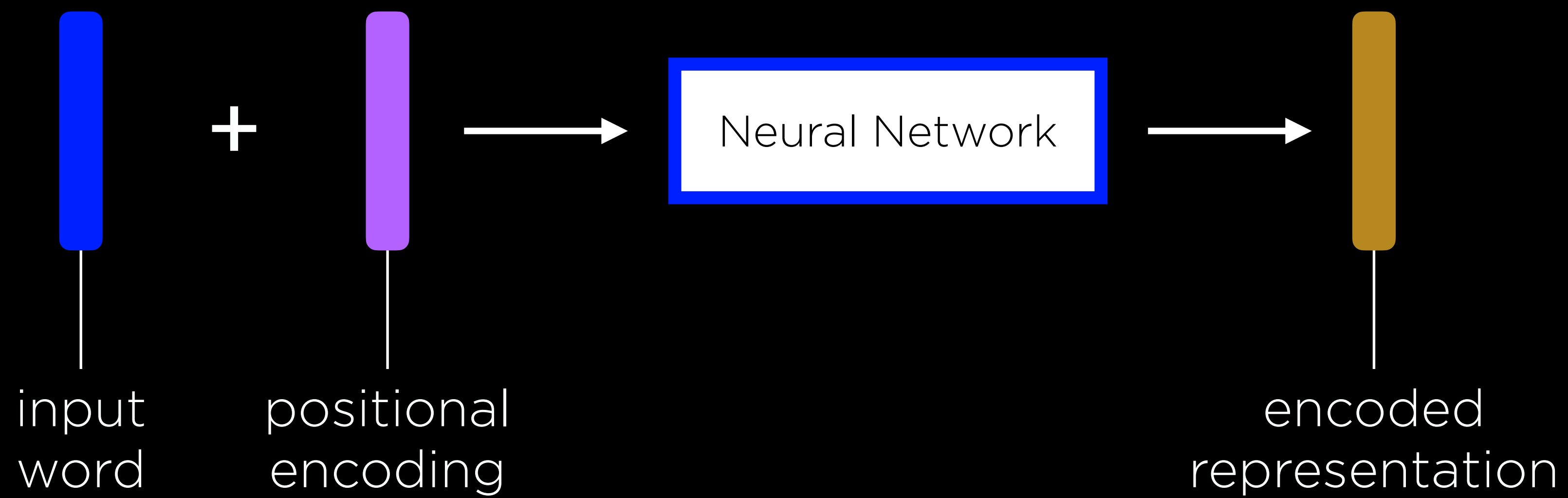
Transformers

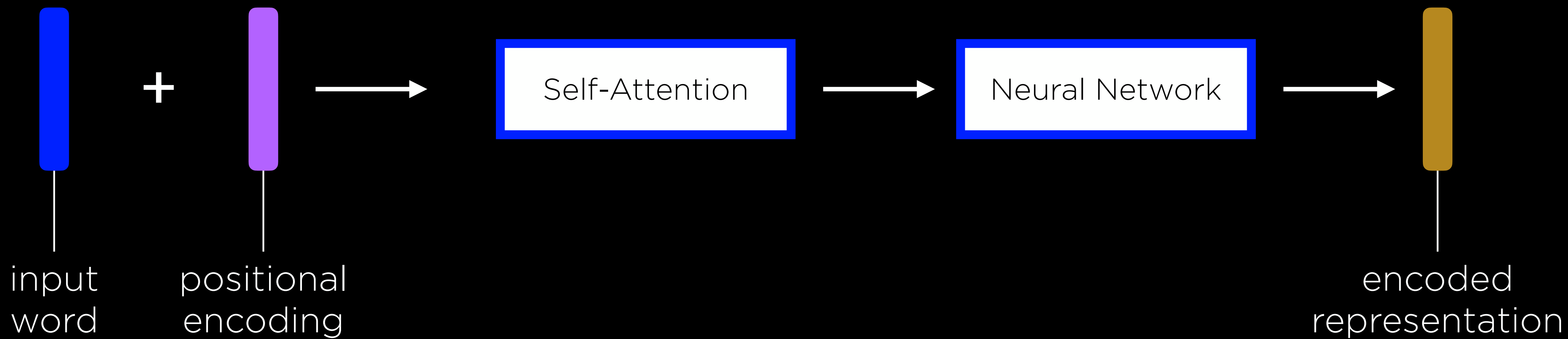


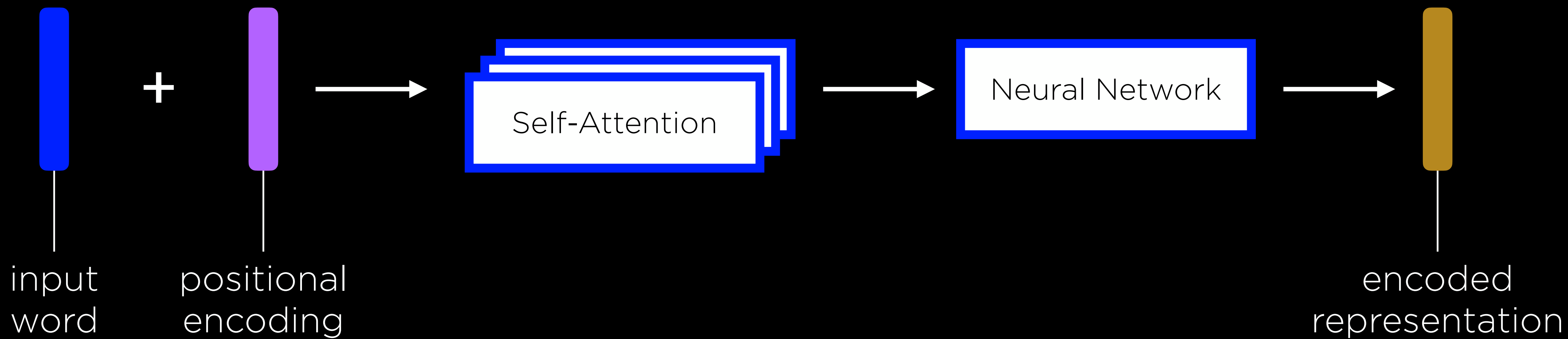
input sequence

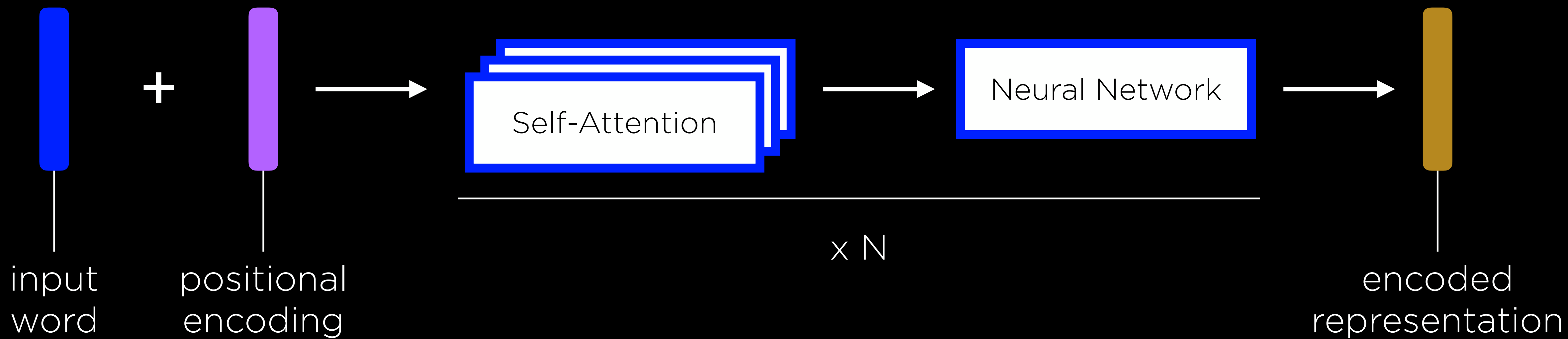




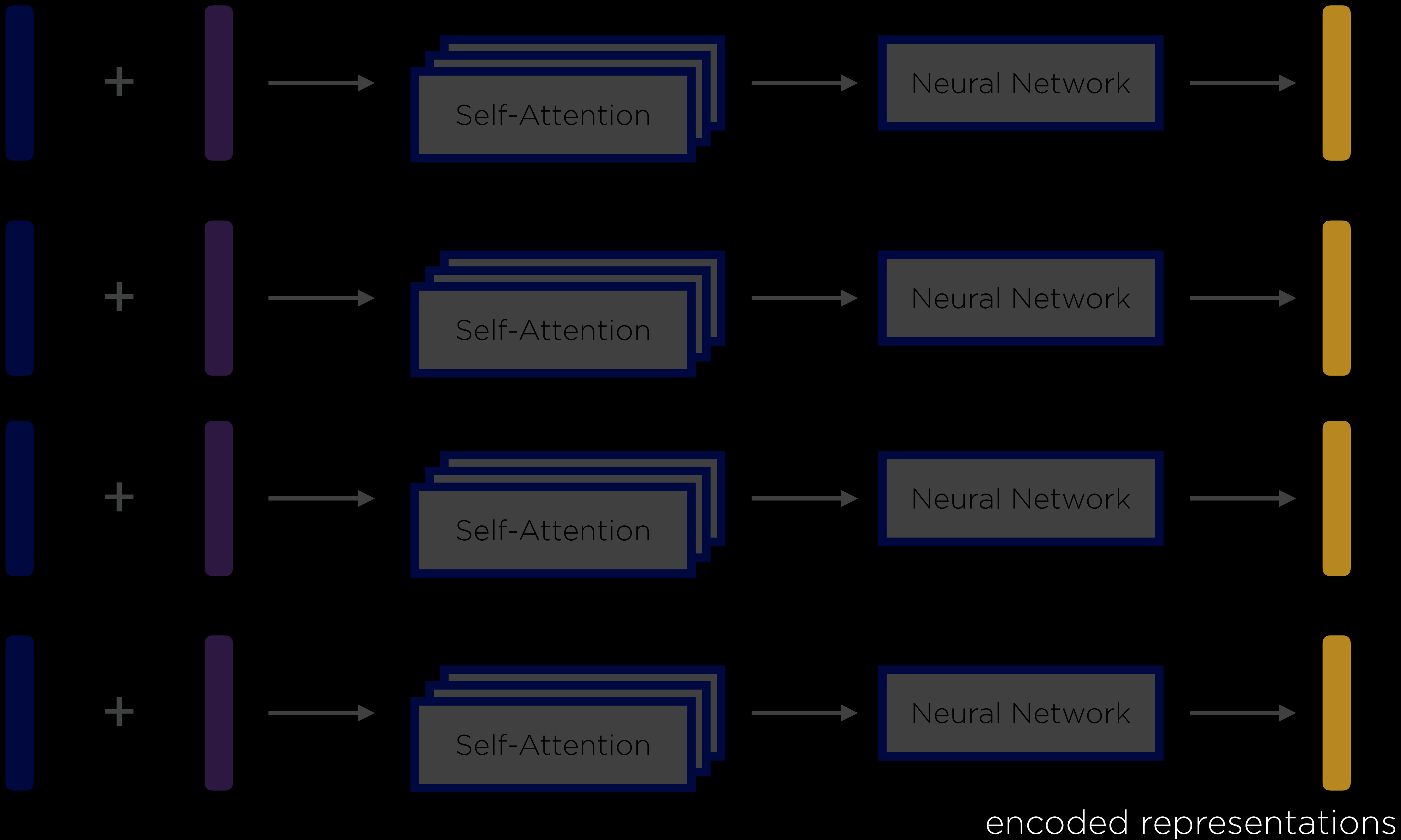


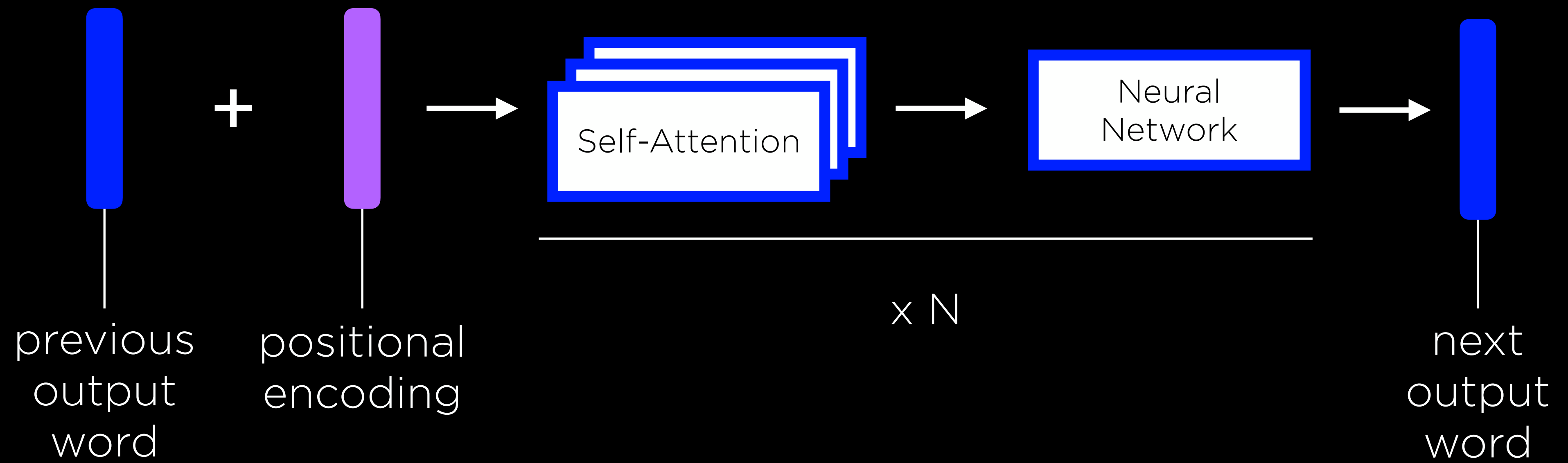


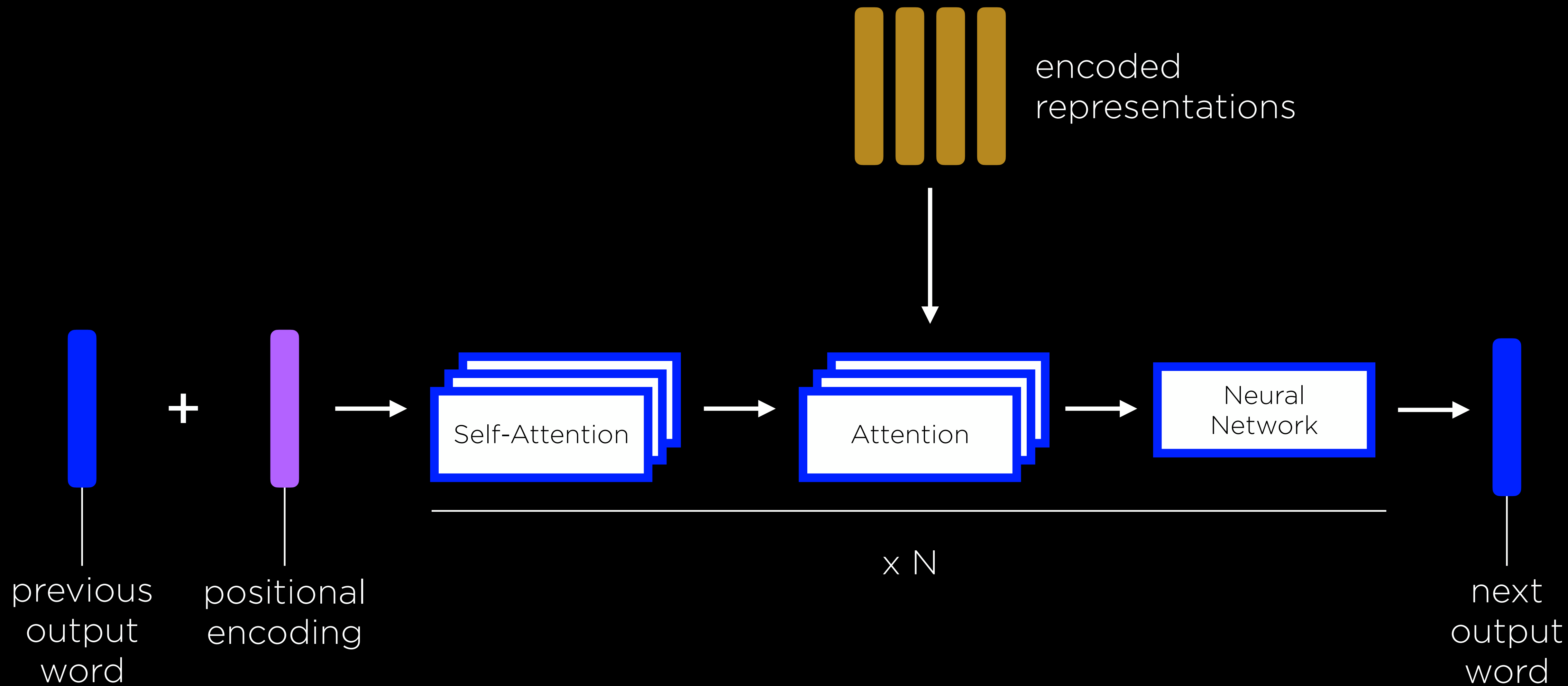


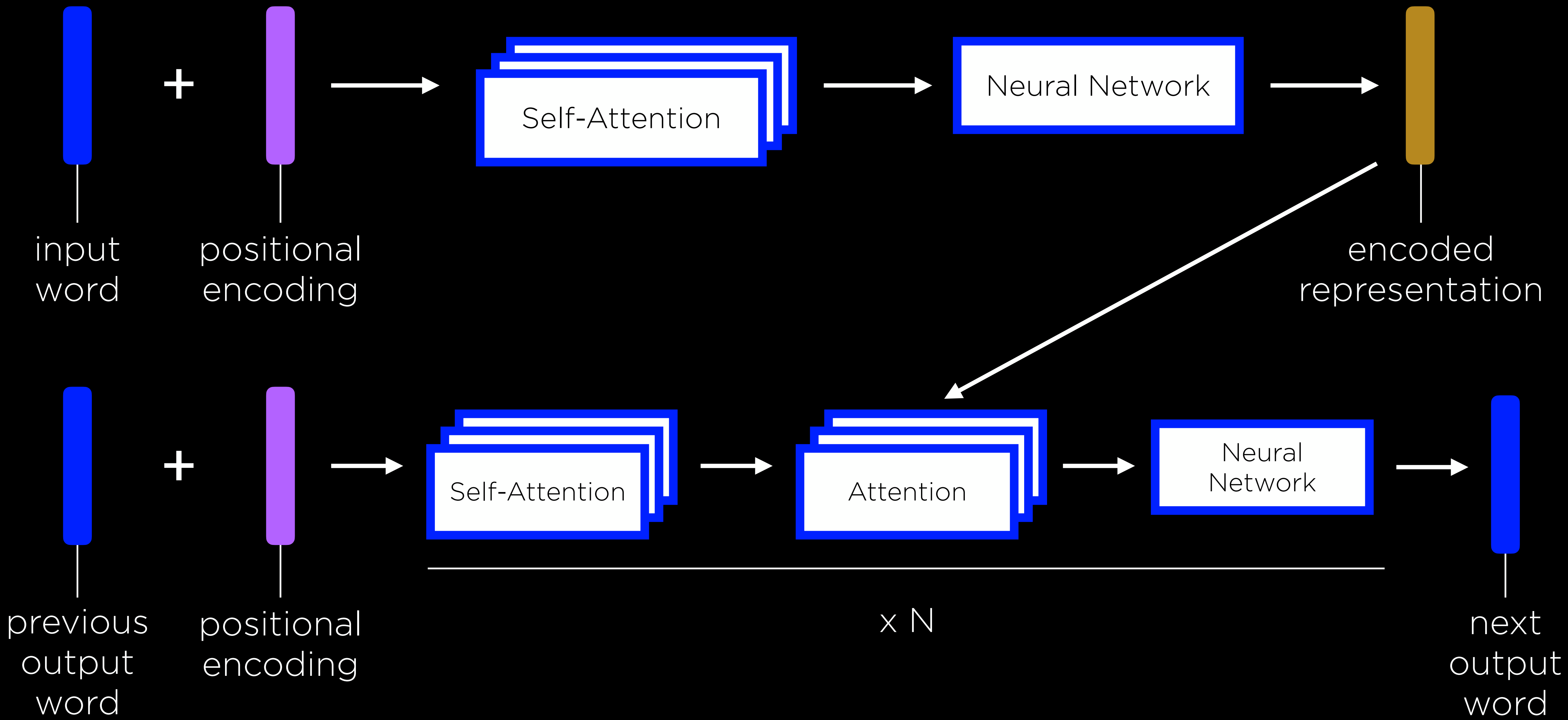








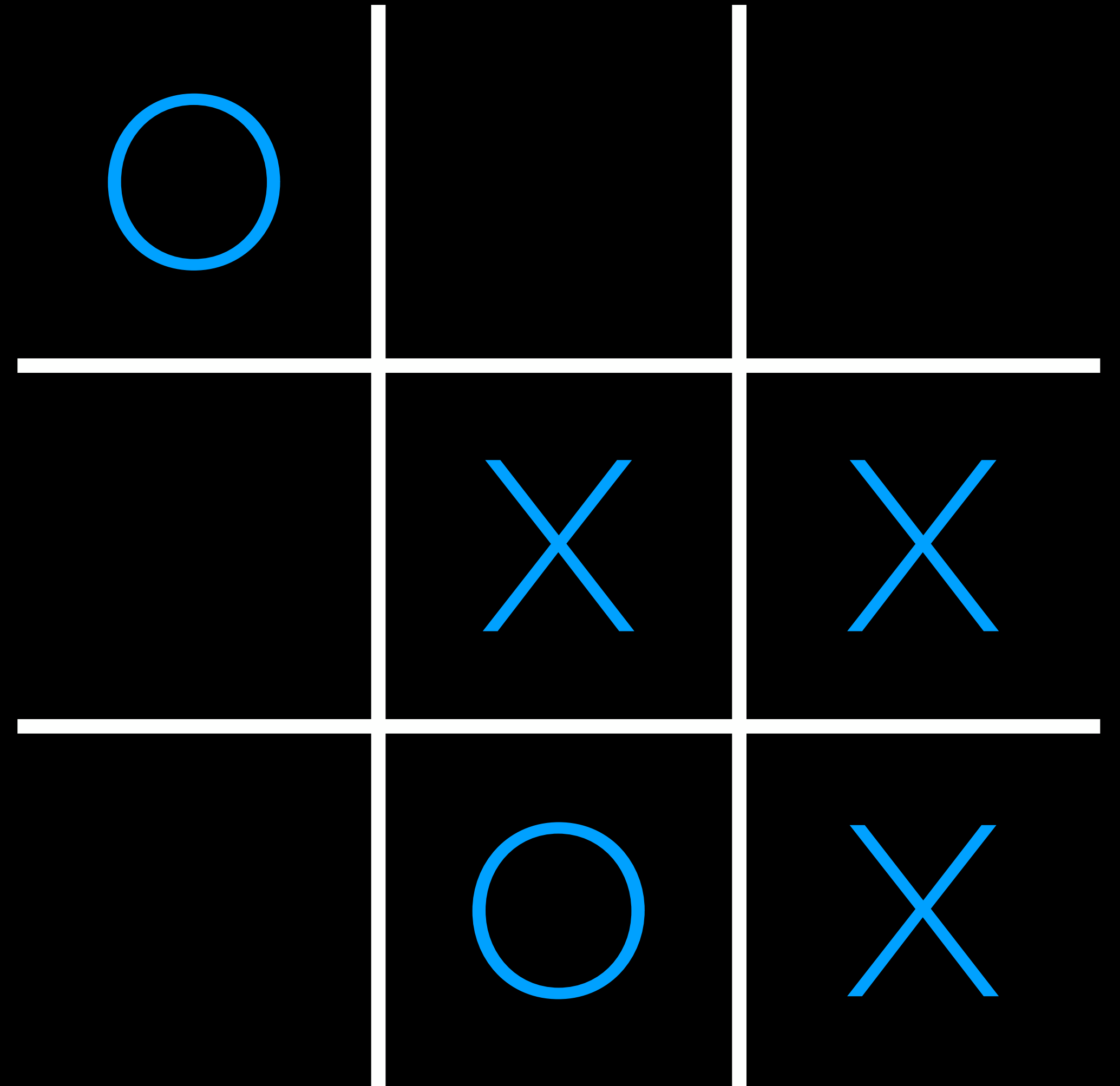




Language

Artificial Intelligence

Search



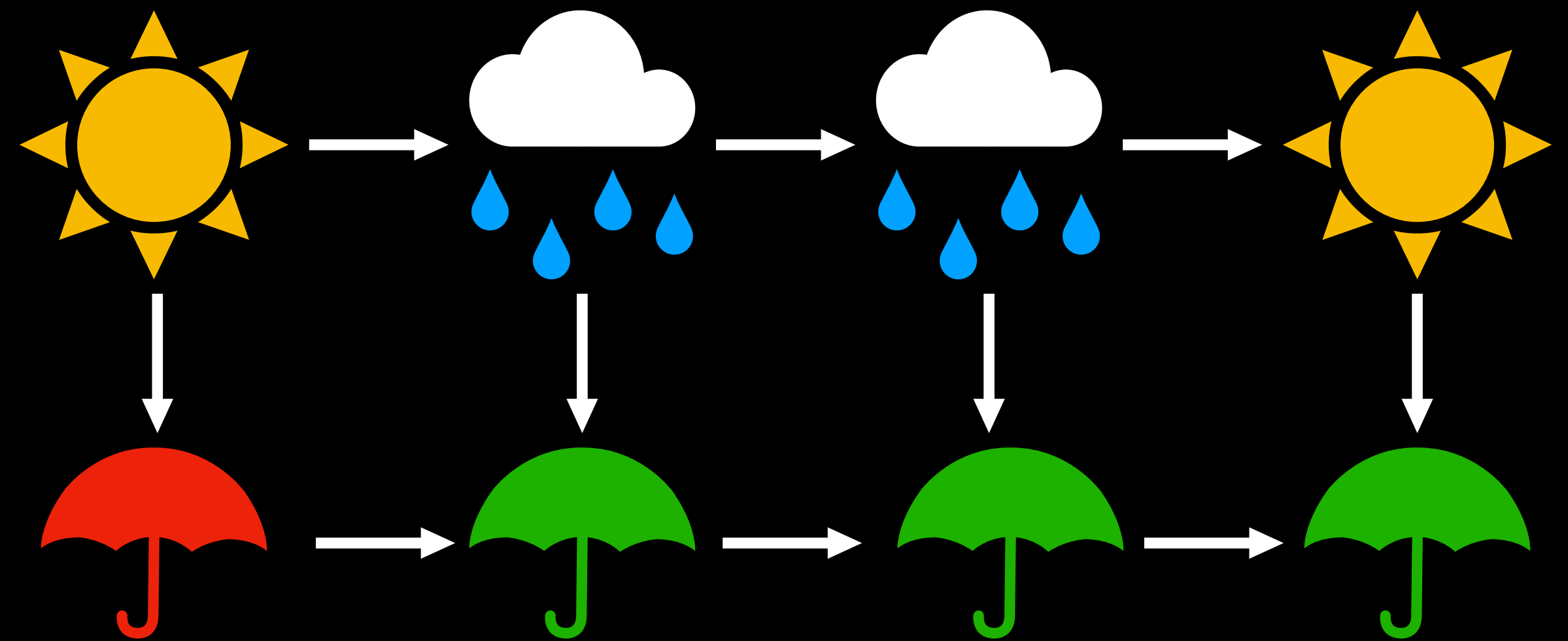
Knowledge

$$P \rightarrow Q$$

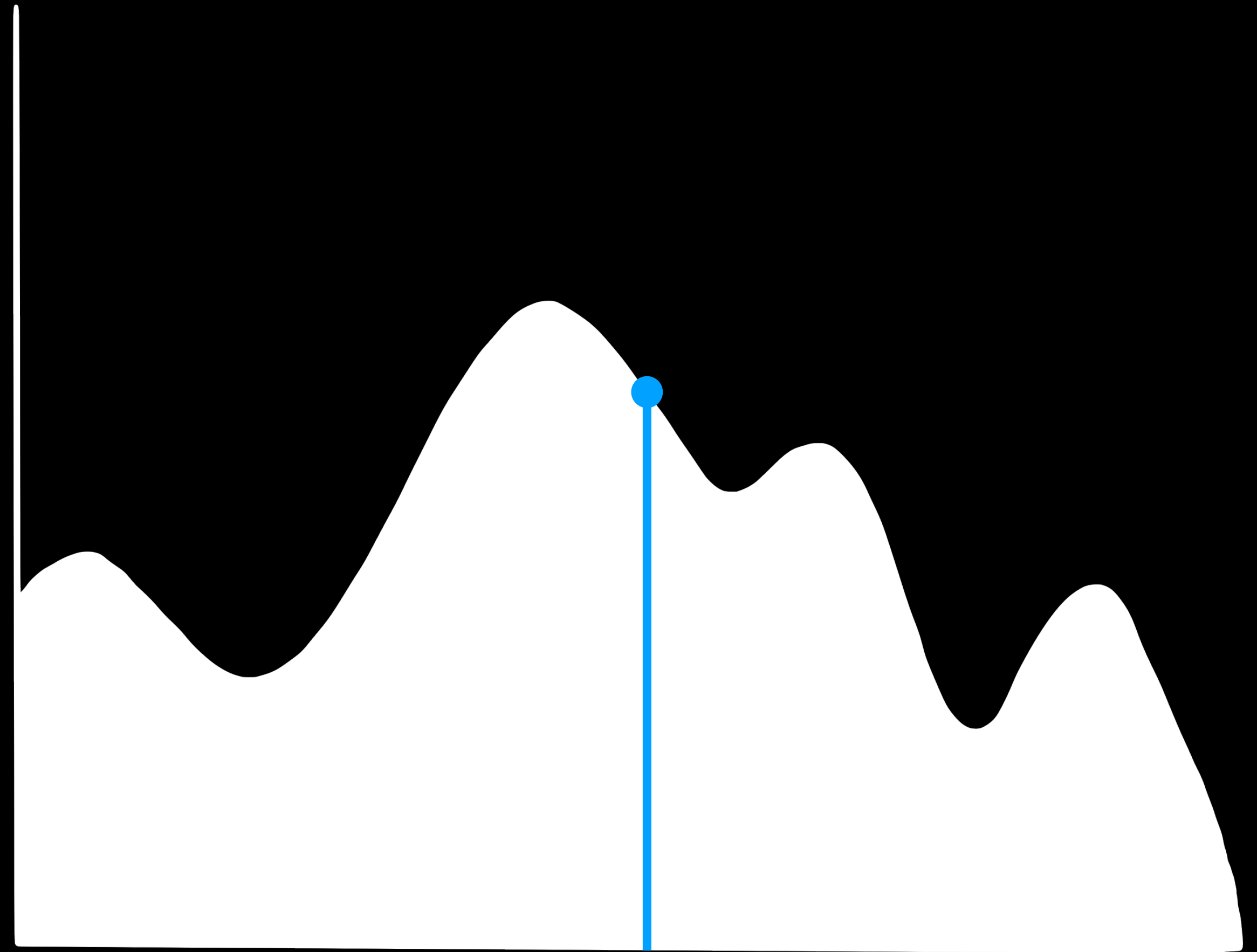
$$P$$

$$Q$$

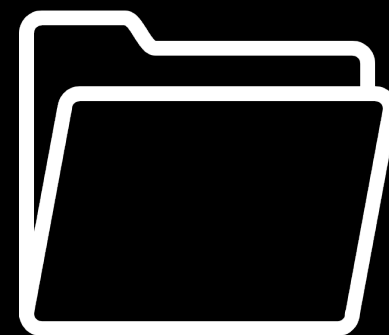
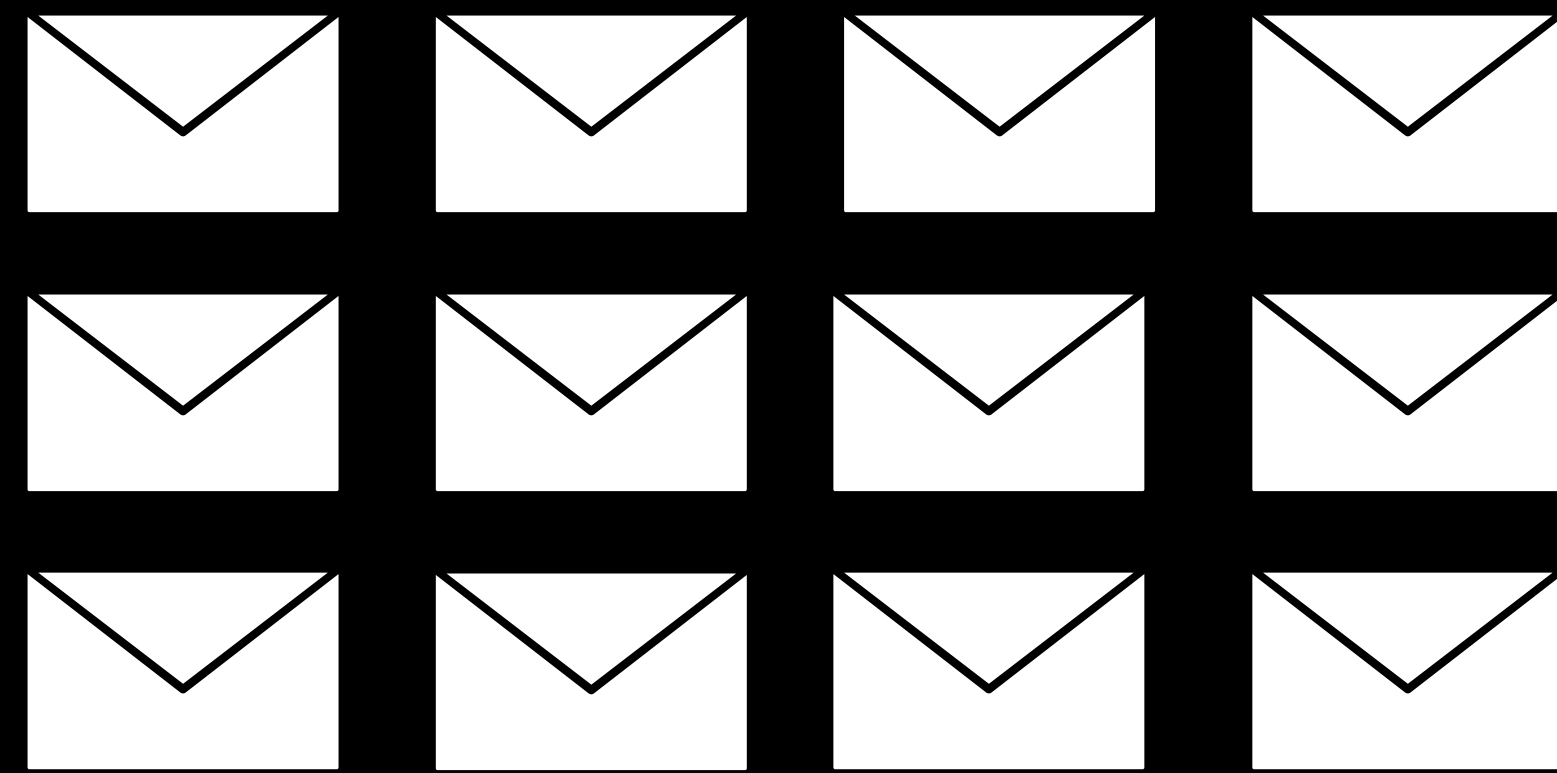
Uncertainty



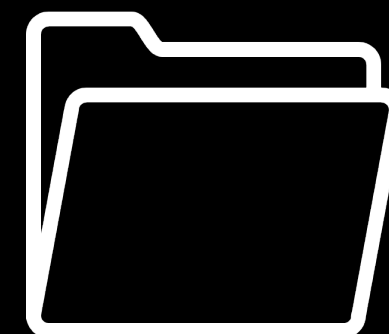
Optimization



Learning

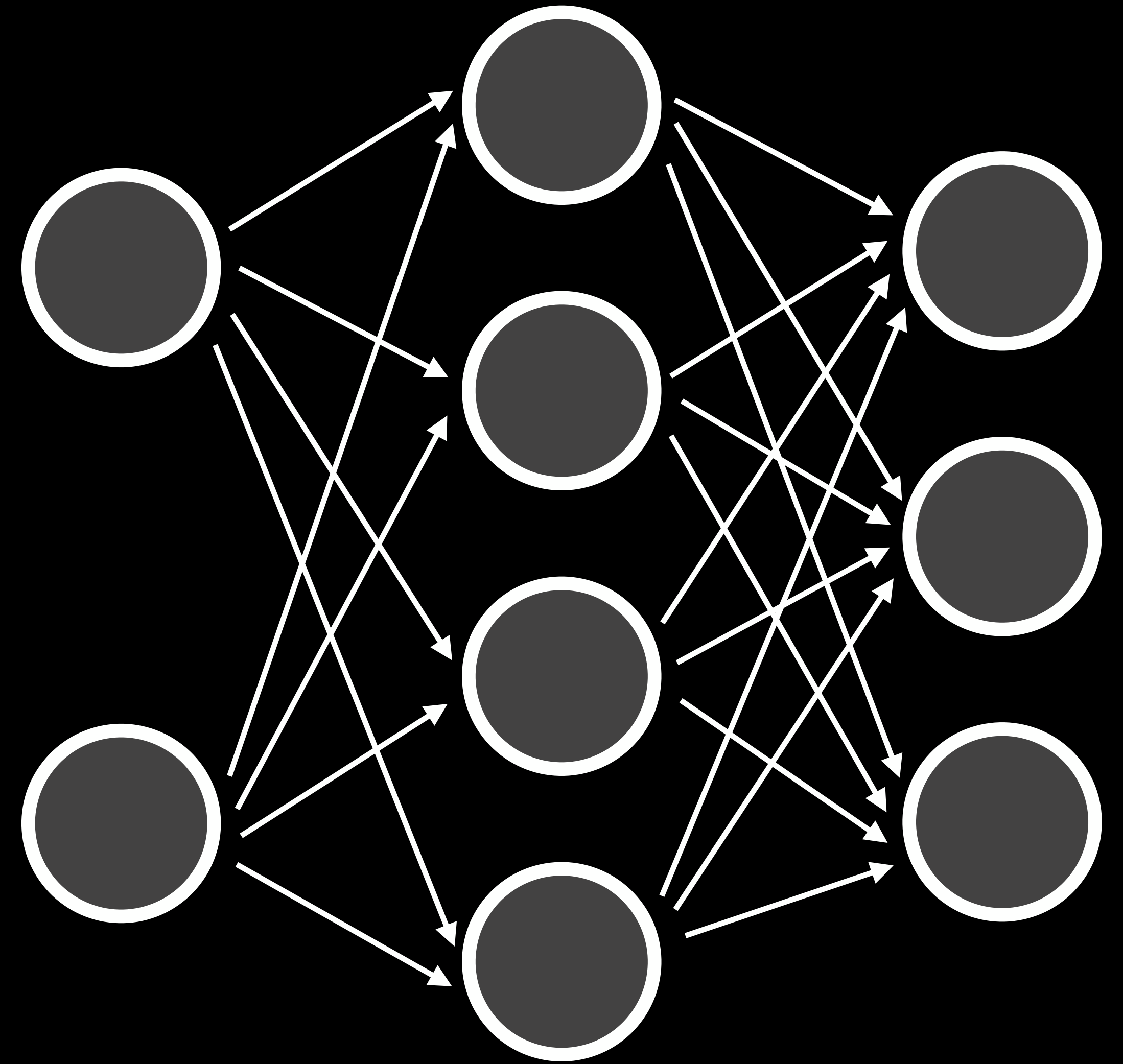


Inbox

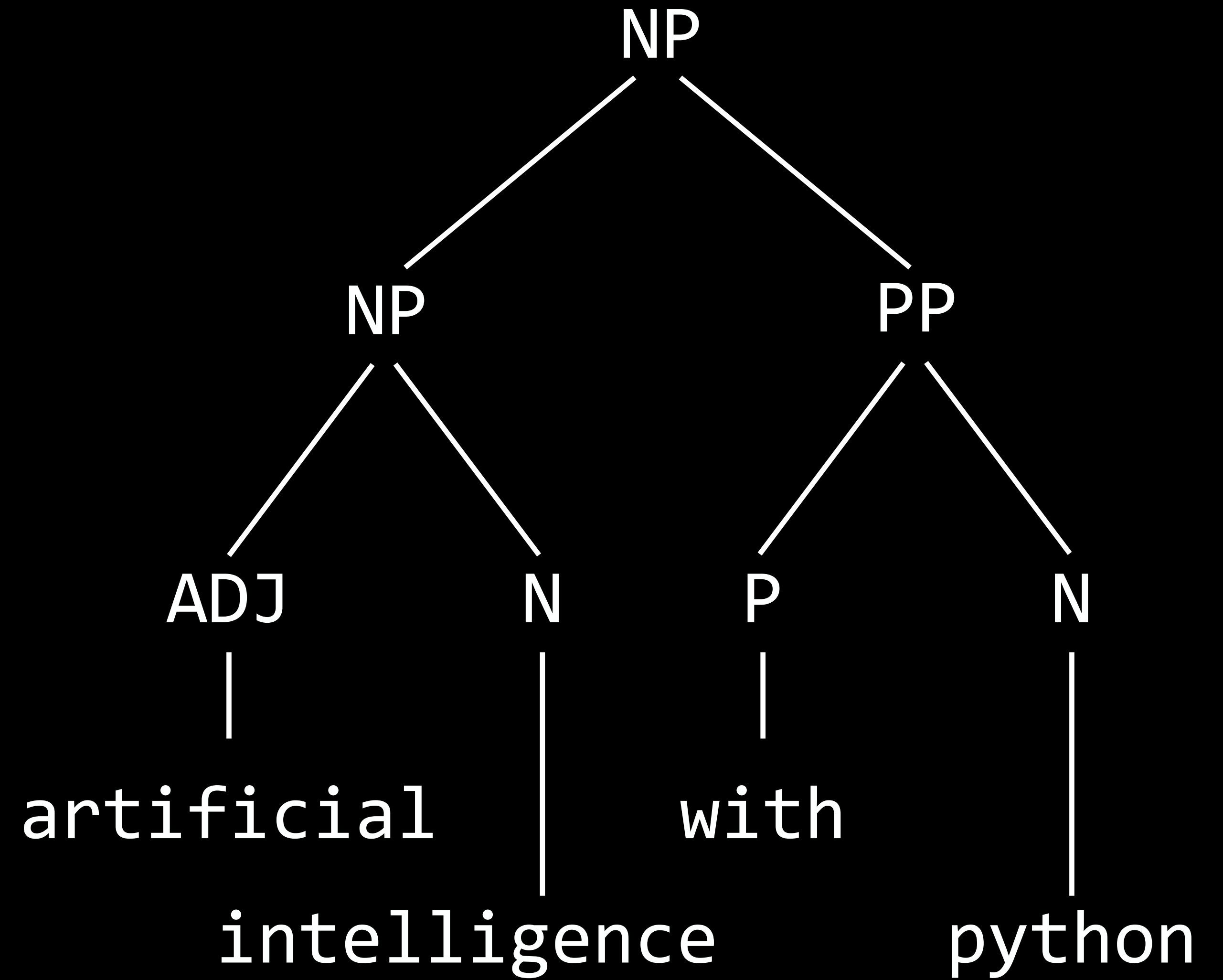


Spam

Neural Networks



Language



Introduction to
Artificial Intelligence
with Python